VEGA-HEREKO ISTANBUL WASTE TO ENERGY PROJECT

Document Prepared by

BIO SOLUTIONS Yenilenebilir Enerji ve Danışmanlık Hizmetleri San. Ve Tic. Ltd. Şti

Project Title	Vega-Hereko Istanbul Waste to Energy Project
Version	01
Date of Issue	19/10/2022
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1 PROJECT DETAILS

1.1 Summary Description of the Project

In order to develop a municipal solid waste management system that will be environmentally friendly and sustainable, Vega-Hereko Istanbul Waste To Energy Project is being implemented by Vega Hereko Enerji Üretim Sanayi ve Ticaret Anonim Şirketi.

The main purpose of the project is to treat MSW with anaerobic digesters and utilize recovered biogas for electricity generation with an **installed capacity**[1] of 6.004 MWe in the first phase. The commissioning was carried out on 03.04.2021 with 6.176 MWm / 6.004 MWe (4*1.544 MWm / 4*1.501MWe) engines. Also, it is planned to reach 18.012 MWe which is license capacity till March 2023. The project activity includes MSW separation, Pre-treatment system, anaerobic digester, RDF Production, composting[2] and electricity generation system.

The projects lifetime is 16 years **and the official license power capacity is 18.012 MWe**, according to the EMRA Generation License[3]. The project has an area of 90,000 square meters with 45,126 square meters building area.

Main objective and aim of the project is to take the separated organic fraction to controlled anaerobic digestion process and recover methane from the organic waste. Residues from the digesters will be sent to separators and then solid residue will be sent to nearest Landfill area and liquid digestate will be fed back to the biomethanization pools before the Composting facility completion date. The methane recovered from digestion is combusted through gas engines and the electricity is fed into the grid. The wastewater is re-used in the digesters, then will be sent to ISTAC wastewater treatment plant near the project area.

The project is located in Karakiraz Village, Şile district city of İstanbul, in Turkey. The project fall into the enforcement of obtaining an Environmental Impact Assessment (EIA), as it can be seen in EIA Positive Certificate[4] and EIA Report[5].

For the purpose of a sustainable waste management in which domestic wastes will be evaluated, it is planned to utilize the Istanbul Site that receives around 730,000 tons/year,

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[1] Please refer to Commissioning Documents.

[2] Composting Production is planned to be installed June,2023.

[3] Please refer to EPDK Generation Licence.

[4] Please refer to EIA Positive Certificate

[5] Please refer to EIA Report

corresponding to an average of 2,000[6] tons per day of municipal waste coming from districts of Istanbul province daily and to generate renewable electric power from biomethanization of organic solid waste and the rehabilitation project was approved by the Ministry of Environment and Urbanization in Turkey.

Whilst providing sustainable development benefits to the host communities and the host country, the project activity reduces greenhouse gas (GHG) emissions mainly by

- preventing GHG emissions, methane in particular, from being emitted directly to the atmosphere from waste at the Istanbul SWDS that would be otherwise left to be decomposed;
- replacing the electricity that would have otherwise been generated by the national grid which is primarily dependent on fossil-fuel-based resources.

The project was commissioned with generation of electricity from April 3rd 2021, which is regarded as the project start date.

This project adopts a crediting period of 7 years. The expected average annual emission reductions are 320,042 tCO2eq/y. Accordingly, the project is expected to generate 2,240,299 tCO2eq emissions reduction throughout its first crediting period.

1.2 Sectoral Scope and Project Type

Sectoral Scope 1: Energy industries (renewable - / non-renewable sources)

Sectoral Scope 13: Waste handling and disposal.

The project is not a grouped project.

1.3 Project Eligibility

The project consists on power generation, methane recovery through controlled anaerobic digestion, RDF Production and Composting, which is eligible under the scope of the Version 4.3 of VCS Standard[7].

1.4 Project Design

The project is not a grouped project.

[6] According to the contract between ISTAC and Vega-Hereko, 2,000 tons/day should arrive. This amount may increase in the future.

[7] https://verra.org/wp-content/uploads/2022/06/VCS-Standard_v4.3.pdf

1.5 Project Proponent

Organization name	BIO SOLUTIONS Yenilenebilir Enerji ve Danışmanlık Hizmetleri Sanayi ve Ticaret Limited Şirketi (LLC.)
Contact person	Adrian CADUFF
Title	Chairperson
Address	Akdeniz Mah. Şehit Fethibey Cad. No: 55 İç Kapı No: 091 Konak, 35210, IZMIR/TURKEY
Telephone	+90 538 338 4626
Email	a.caduff@biosolutions.com.tr

Organization name	Vega Hereko Enerji Üretim Sanayi ve Ticaret Anonim Şirketi
Contact person	Tuğba Kırer
Title	Board Member
Address	Ostim OSB Mah. 1427 Cad. No:1A Yenimahalle / ANKARA
Telephone	+90 532 405 23 69
Email	tkirer@vegaturkiye.com

1.6 Other Entities Involved in the Project

N/A

1.7 Ownership

Both the attached approval of Environmental Impact Assessment (EIA) issued by the Ministry of Environment and Urbanization, Directorate General of Environmental Management in Turkey and the attached the Electricity Generation License (EGL) issued by the Energy Market Regulatory Authority (EMRA) in Turkey on February 11, 2021 established the project ownership, Waste to Energy Plant, to Vega Hereko Enerji Üretim Sanayi ve Ticaret Anonim Şirketi. These official documents are accordingly establishing the property and contractual right both in the plant and equipment that generate GHG emission reductions.

1.8 Project Start Date

The project start date is 03/04/2021, as the operation start date with generation of electricity commissioning.

1.9 Project Crediting Period

Each crediting period is 7 years. The crediting period will be renewed two times more. Therefore, total length of the crediting period will be 21 years. The first crediting period starts in 03/04/2021 and finishes 02/04/2028.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	
Large project	√

Table 1. Estimated GHG emission Reductions or removals (tCO2e)

Year	Estimated GHG emission reductions or removals (tCO_2e)
1	146,065
2	216,684
3	280,145
4	330,706
5	380,431
6	424,016
7	462,254
Total estimated ERs	2,240,299
Total number of crediting years	7
Average annual ERs	320,042

1.11 Description of the Project Activity

For the purpose of sustainable waste management, it was planned to rehabilitate the Vega-Hereko Istanbul Area and with the completion of the rehabilitation project by phases. One project activity is going to be producing high quality RDF from light fraction and the second one is the methane recovery through controlled anaerobic digestion to produce electricity. Third one is going to be composting with the residues from the digesters.

The Municipal solid waste (MSW), goes through mechanical sorting to separate most of the organic fraction of the waste. Around 50% of the waste which is rich organic waste goes to a Biodigester. Finally, the solid output of the Biogas is going to be sent to composting, then will be outward transported and will not be stored under anaerobic conditions.

The gas for the gas engines is delivered together from the biogas plant. This is estimated to be around 730 m³/h for each engine from the biogas plant and the biogas plant a methane level of about 52% - 55% is available due to the high content of added subcutaneous fat to the biogas process.

The project capacity is 18.012 MWe. The waste received is around 730,000 tons per year, corresponding to an average of 2,000/t day. The project will collect the wastes regularly and will help to decrease irregular distribution areas in Turkey. Waste amount is expected to rise by 0,4%[8] per year in coming years.

The mechanical sorting of the waste with 9,000 tons/day capacity is in operation since project start date. There is no distribution of natural gas through pipelines nor trucks to end users.

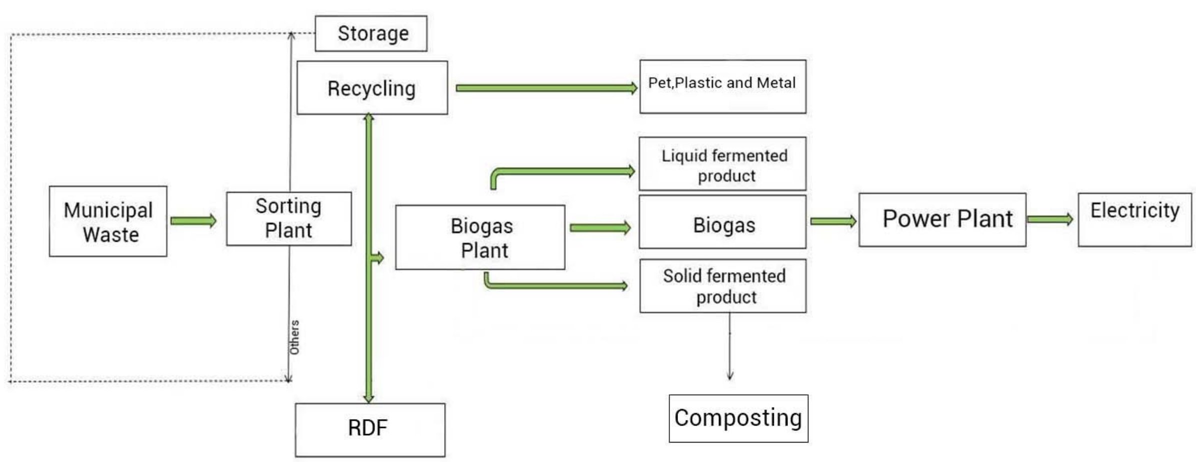

Figure 1. Flow Process Chart for the Vega-Hereko Istanbul Waste to Energy Project

[8] Please refer the feasibility report, page 24.

The different units conforming the whole Vega-Hereko Istanbul Waste to Energy projects are presented in the following sections:

a) Sorting System:

With the system with a daily capacity of 9,000 tons per day, domestic waste coming to the facility is separated to achieve more efficient results.

After the separation of 50% of domestic waste per day, high organics are sent to the bio fermenter and low organics are going to be sent to the RDF system.

Figure 2. Sorting System from outside

b) Biogas Production Unit:

The extracted organic waste is sent to Biogas Production Unit for anaerobic digestion. The below figure illustrates the digester to be used in the biogas production unit;

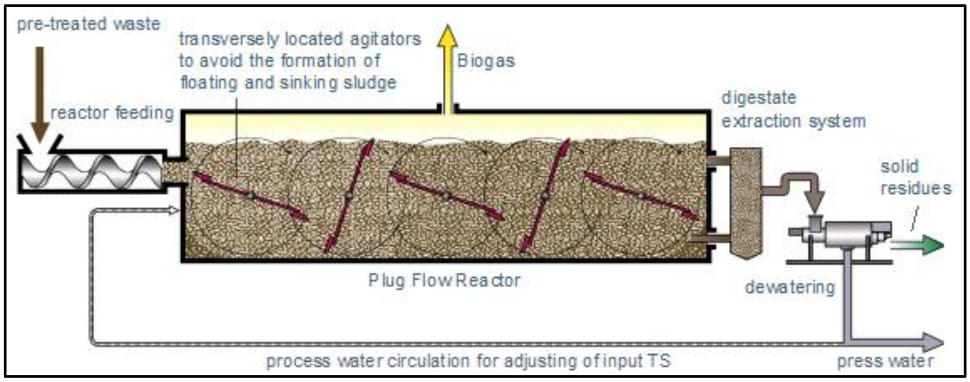

Figure 3. Flow Chart of Digester

c) Flaring Unit:

The facility has a flaring unit to prevent the release of the landfill gas in case of any breakdown or failure in the facility or when the facility is out of service. The landfill gas would be flared in such conditions. There are 2 storage balloons with capacity of 10,000 m3 to ensure that constant volume is maintained.

d) Gas Cooling and Cleaning Area:

In the gas cleaning unit, landfill gas is processed and cleared of particles and moisture inside. The condensates (water) are sent to the water treatment plant of the ISTAC together with the leakage water.

e) Power Generation in Internal Combustion Engines:

The gas utilization unit will be fitted with 12 CHP plants (12 x (Jenbacher 1.544 MWm / 1.501 MWe)). The gas for the CHP-plants will be received from the biogas plant. The generated power is fed into the Grid and it is further delivered to the consumers through power transmission lines.

Since the internal combustion engines require less gas flow rate and are easier to switch on and off when compared to the gas turbines, they are more suitable for landfill gas power plants. Mostly all of the landfill gas - biogas power plants in our country consist of internal combustion engines.

- **Internal Combustion Gas engines (ICE)**

Internal combustion engines are the most suitable method for the utilization in landfill gas power plants.

The engines installed in the project are Genset JGS 420 GS-B.L characterized by a high-power density and efficiency.

Table 2. Technical specifications Jenbacher Genset JGS 420 GS-B.L[9]

Technical specifications	
Electrical output	1501 kW
Electrical Efficiency	Up to 41.9%
Voltage	400V
Gas Volume	796 Nm3/h
Starter Motor Output	13 kW
Fuel Type	Flexible
	

[9] Please refer the Techinal Specifications of Gas Engine

- **Technology Units**

The technology employed will be the improvement of landfill gas collection and flaring through the installation of an active recovery system composed by:

Table 3. Machine and Equipment Information

Number	MACHINE AND EQUIPMENT (Type and Technical Properties)	Capacity (Information about power or more)
3	Blower	AIRCOO HG-100
6	Blower	HG-100
12	Gas Engine	JENBACHER J420 GS-B25 1501 kW
1	Generator (diesel power unit)	GENSET MPM 15/400 I KA 10 kVa
1	Generator (diesel power unit)	EMSA E SD EM 0290 290 kVA
1	Generator (diesel power unit)	EMSA E SD EM 0290 290 kVA
1	Transformer	1400 kVA
4	Transformer	1600 kVA
13	Transformer	2000 kVA
1	Windshifter Filtre	MOLDOW 070SBS-316EX
1	Flare	1000 m³/h 1000 °C
1	Mechanical separation system	9,000 tons/day

1.12 Project Location

The project was implemented over an area of roughly 156.600 acres of area in boundary of Vega-Hereko Istanbul Waste to Energy Facility in Karakiraz Village, Istanbul Site, İstanbul Province, Şile District. The coordinates of the project location are longitude of 29°21'57.07"E and latitude of 41°8'32.33"N.

Table 4. System Coordinates

Point	(Geographical-WGS-84)-(Degree, Fraction)	
	Latitude	Longitude

1	41.1443273	29.3682056
2	41.1431345	29.3697115
3	41.1422658	29.3699066
4	41.1421241	29.3671612
5	41.1420609	29.3667362
6	41.141712	29.3657404
7	41.1407105	29.3640218
8	41.1412084	29.3629611
9	41.140775	29.3623558
10	41.1422529	29.3605188
11	41.143679	29.3624988
12	41.1429405	29.3642476
13	41.1434161	29.3660877

Figure 4. Location of the Istanbul Province

Figure 5. Location of the Vega-Hereko Istanbul Project. Source: Google Earth

Figure 6. Photograph of the Vega-Hereko Istanbul WTE Project

1.13 Conditions Prior to Project Initiation

The scenario existing prior to the start of the implementation of the project activity is:

Before the proposed project activity, the facility was only engaged in the production of RDF and the recyclable materials. In addition, the quantity of electricity produced using LFG would have been produced by power plants connected to the grid in Turkey which is heavily dependent on fossil fuel resources. The baseline scenario is the same as the scenario existing prior to the start of implementation of the project activity.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project has obtained the approval of the Electricity Generation License (EGL) issued by the Energy Market Regulatory Authority (EMRA) in Turkey on February 11th, 2021. All necessary permits were obtained under the current legislation (please refer to EIA documents).

The two of main regulatory approvals for such project types to prove that they are in compliance with relevant laws, statues and regulatory frameworks in Turkey are the Environmental Impact Assessment and the Electricity Generation License whose obtainments are required for projects being in compliance with other relevant legal requirements. *Environmental Permit and License application will be made within the scope of Environmental Permits and Licenses Regulation and Communique on Mechanical Sorting and Biomethanization Facilities and Fermented Product Management*.

All necessary permits were obtained under the current legislation (please refer to EIA documents). Hence, the project is in compliance with laws, status and other regulatory frameworks.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not been registered or is seeking registration under any other GHG programs.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading. (Please refer to declaration of non-participation under other GHG programs)

1.16.2 Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related credit or renewable energy certificates.

1.17 Sustainable Development Contributions

The Project will contribute to sustainable development in the following ways:

• Increasing labor demand of skilled labor for the fabrication, installation, operation and maintenance of the methane recovery and electricity generation system and thus, contributing to the sustainable economic growth of the region,

• Generating and dispatching electricity from a renewable and sustainable energy source to a grid nowadays reliant on fossil,

• Contributing to the climate change fight by reducing CH_4 emissions.

• Constituting a new, clean and efficient technology model for the disposal and handling of waste

• Improving air quality (i.e. by reducing odor) and therefore having positive effects on the local environment.

Nevertheless, it is still possible to note that the project will make positive contributions to at least three Sustainable Development Goals (SDGs). These are:

- **SDG Goal 7: Affordable and Clean Energy**

The proposed Project is a waste to power project that will generate renewable energy by capturing methane from municipal waste and utilizing it to produce thermal and electric energy through gas engine systems. By supplying renewable energy generated at the plant to the national grid, the proposed Project will contribute to increasing the share of renewable energy in the global energy mix and the proportion of the population with primary reliance on clean fuels and technology.

- **SDG Goal 8: Decent Work and Economic Growth**

The demand for food and electric energy is rapidly increasing in Turkey for various reasons, such as industrialization, urbanization, economic development, and population growth. The country's external dependence on agricultural products has intensified because the increasing demand cannot be met by a decreasing domestic agricultural production capacity which is due to several reasons, such as shrinkage of agricultural land, increasing migration to urban spaces from rural areas where agricultural production is densely located, and dramatic increase in the cost of inputs for agricultural production. Therewithal, Turkey, which cannot meet its increasing electricity demand due to the deprivation of conventional resources used to generate energy, such as coal, oil, and natural gas, has also become foreign-dependent on energy. Through its implementation, the proposed project activity will contribute to reducing Turkey's foreign dependency by generating renewable energy out of municipal wastes. In addition, as an LFG-based renewable energy technology implementation, the proposed project activity will achieve higher levels of economic productivity; hence, increasing the annual growth rate of real GDP per employed person. Moreover, it will increase the region's employment capacity while decreasing the unemployment rate.

- **SDG Goal 13: Climate Action**

The proposed project activity will reduce GHG emissions by capturing and utilizing methane, one of the most potent GHGs triggering climate change. It is estimated that the average annual emission reduction that the proposed project will generate is around 320,042 tCO2eq.

1.18 Additional Information Relevant to the Project

Leakage Management

There are two possible leakage sources, one is the liquid leakage of the Biodigester that is captured and sent to the water treatment facility of the municipality, and second, the CH4 emissions that are sent to flare in case of an emergency or an unlikely case of run out of the gas engines.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

N/A

2 SAFEGUARDS

2.1 No Net Harm

As mentioned in section 1.14 COMPLIANCE WITH LAWS, STATUTES AND OTHER REGULATORY FRAMEWORKS the project meets all necessary regulatory requirements such as socialization of the project, solid waste, water pollution, noise control and air quality protection, it is therefore possible to claim that there is no net harm linked to the project by supporting reports.

2.2 Local Stakeholder Consultation

The project's local stakeholder consultation meeting took place on May 26th, 2022.

Invitation has been sent to the stakeholders between 5-6 May 2022 via, newspaper, Email, notice at public place and personal invitations.

A few invitation samples are here:

11 Mayıs 2022 — ŞİLE GÜNDEM 7

ŞAKAYIK FESTİVALİ MUHTEŞEM OLDU!

MUHTEŞEM BAYRAM!!!

Maden Ocağı Kaynaklı Oluşan Göçük Yolu Kapattırdı. İYİ Partililer Vatandaşın Yanında Boy Gösterdi. Halkın Mesajı İse Net:

"BÖLGEMİZDE VAHŞİ MADENCİLİK İSTEMİYORUZ!"

ZAYİ

ŞİLE GÜNDEM

VEGA HEREKO ENERJİ ÜRETİM TESİSİ
YEREL PAYDAŞ BİLGİLENDİRME TOPLANTISI

Vega Hereko Enerji Üretim Sanayi ve Ticaret Anonim Şirketi olarak, İstanbul İli, Şile İlçesi ve Karakiraz Mahallesi sınırları içerisinde işlettiğimiz enerji üretim tesisinde; küresel iklim krizi ile mücadeleye, yenilenebilir enerji üretiminin artışına, enerji tüketiminde dışa bağımlılığının azaltılmasına katkı sağlamayı, kentsel atıkları değerlendirerek yenilenebilir enerji üretmeyi ve bölgede oluşan çevresel etkileri azaltmayı hedeflemekteyiz.

Firmamız, projenin yaratmış olduğu sera gazı emisyon azaltımının uluslararası seviyede belgelendirilmesi amacıyla VCS Gönüllü Emisyon Azaltım sertifikalandırma sürecini başlatmış bulunmaktadır.

Bu kapsamda değerli görüşlerinizi almak ve sizleri bilgilendirmek için düzenlenecek olan, **Yerel Paydaş Bilgilendirme Toplantısı**'nda siz İstanbul halkını, kurum ve kuruluşları bu etkinlikte görmekten memnuniyet duyacağımızı belirtmek isteriz.

Toplantıya katılımlarınızın projemiz için önemli olduğunu tekrar belirtir, konu ile ilgili olarak bilgi ve gereğini arz ederiz.

Toplantı Tarihi: 26.05.2022
Toplantı Saati: 14:00
Toplantı Yeri: Çevrimiçi yapılacaktır.
Kayıt Linki: https://t.ly/vegahereko

Telefon kameranız ile aşağıdaki görseli okutarak toplantı kayıt linkine ulaşabilirsiniz.

TOPLANTI PROGRAMI:

14:00-14:05: AÇILIŞ VE TAKDİM

14:05-14:20: PROJE TANITIMI

14:20-14:40: İKLİM DEĞİŞİKLİĞİ, SÜRDÜRÜLEBİLİR KALKINMA, KARBON DENKLEŞTİRME MEKANİZMALARI VE KARBON SERTİFİKALANDIRMA SÜREÇLERİ

14:40-15:00: KATILIMCILARDAN GELEN SORULARIN CEVAPLANDIRILMASI

15:00-15:15: SÜREKLİ GERİ BİLDİRİM SÜRECİNİN TANITILMASI

15:15-15:30: SÜRDÜRÜLEBİLİR KALKINMA ADIMLARININ TANITIMI ve DEĞERLENDİRMESİ

15:30-15:45: SÜRDÜRÜLEBİLİR KALKINMA TAKİP MEKANİZMALARI ve KAPANIŞ

Toplantıya katılamayacak olup görüşlerini bildirmek isteyen ilgililerin ilayda.onaran@ch4climate.com adresi ile iletişime geçmesini rica ederiz.

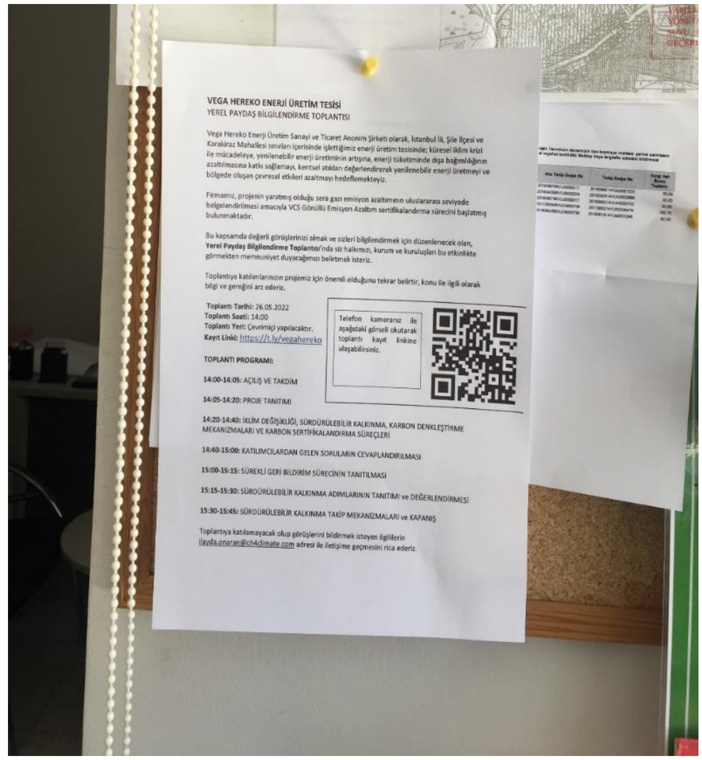

Figure 7. Photographs of Local Stakeholder Meetings Invitation Letters

The Agenda which was shared with the local stakeholders before and during the LSC meeting can be seen below:

Date of the Meeting: 26.05.2022

Time: 14:00 - 15.45

Location: Online Webinar, Microsoft Teams

Agenda:

14:00	14:05	- Opening of the meeting & Introduction
14:05	14:20	- Explanation of the project
14:20	14:40	- Brief introduction to climate change, sustainable development, carbon offsetting mechanisms and certification processes
14:40	15:00	- Q&A Session about the project
15:00	15:15	- Discussion of continuous input/grievance mechanism

15:15 15:30   - Sustainable Development Contributions (SDC) explanation

15:30 15:45   - Discussion on monitoring SD & Closure of the meeting

During the 5th of May, announcements were carried out to reach the identified stakeholders. First of all, invitation letters were posted in the bakery, supermarket, bus stop and restaurant, which are the most frequently visited spots in the Şile region, to inform the local people. During the hanging of the invitation letters, all citizens found in the stores were informed by the project officer in order to register and participate in the meeting.

Minutes of the LSC Meeting

The local stakeholder information meeting was held on 26th May 2022 as agreed. At the beginning of the meeting, it was explained in detail to the participants of the online meeting, which was held due to the Covid-19 outbreak, how questions about the project would be collected during the meeting and how they could give feedback.

The demographic information of the participants was received during the registration for the online meeting. On the registration page of the meeting held over Microsoft Teams, non-technical information about the project is also summarized and explained.

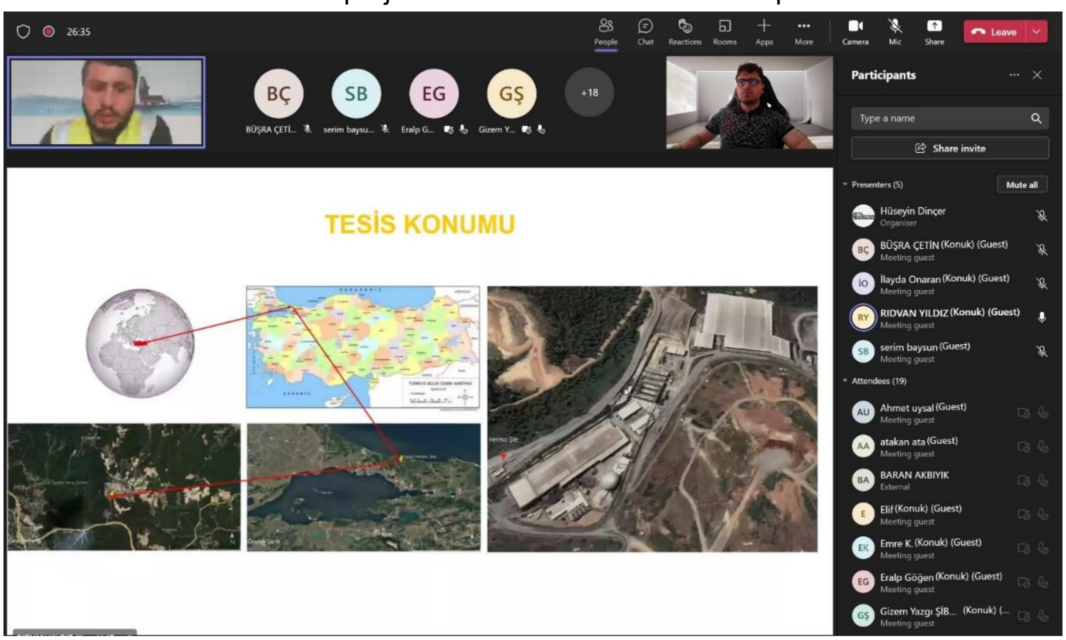

Figure 8. Photographs of Local Stakeholder Meeting

The contact information of the project developer was shared in both electronical and printed invitation letters (newspaper, invitation, announcement, etc.) so that those want to provide feedback could communicate directly. The project personnel stated his corporate e-mail address and phone number during the meeting. Public comment period after project listing was revealed during the LSC meeting. In addition, feedback forms and the business card of the project manager were also given to the mukhtar of the neighborhood to enable stakeholders to provide feedback.

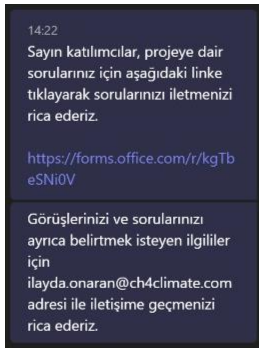

Figure 9. Link for attendees questions

Also, considering the questions and feedback of the participants attended the meeting, there is no negative opinion encountered regarding the facility. Taken actions are explained about related questions.

2.3 Environmental Impact

The environmental matters in Turkey are jurisdiction of the Ministry of Environment and Urbanization. The Law defines among others the general requirements of the Environmental Impact Assessment (EIA) states specific requirements according to the type of project, location, size and other characteristics.

However, all permits related to the regulations of the Environmental Law No. 2872 were obtained such as; Solid Waste Control Regulation, which came into force by being published in the Official Gazette dated 14.03.1991 and numbered 20814, Noise Control Regulation (entered into force by being published in the Official Gazette dated 11.12.1986 and numbered 19308), Water Pollution Control Regulation (entered into force by being published in the Official Gazette dated 04.09.1988 and numbered 19919) and it is necessary to comply with the conditions specified in the Regulation on the Protection of Air Quality, which came into force after being published in the Official Gazette dated 2.11.1986 and numbered 19269.

Considering the fact that the project activity already obtained the mandatory regulation, it is therefore possible to claim that there is no net environmental impact linked to the project.

2.4 Public Comments

After project listing, the VCS/Verra project Vega-Hereko Istanbul Waste to Energy Project will be open for 30 days of public consultation.

2.5 AFOLU-Specific Safeguards

Since the project is a non-AFOLU project, this section is not required.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The approved consolidated methodology applied in the project activity is ACM0022 Alternative waste treatment processes (Version 03.0).

This methodology also refers to the latest approved version of the following:

- TOOL02: “Combined tool to identify the baseline scenario and demonstrate additionality” (version 07.0);
- TOOL04: “Emissions from solid waste disposal sites” (version 08.0);
- TOOL05: “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0);
- TOOL06: “Project emissions from flaring” (version 03.0);
- TOOL07: “Tool to calculate the emission factor for an electricity system” (version 07.0);
- TOOL08: “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” (version 03.0);
- TOOL14: “Project and leakage emissions from anaerobic digesters” (version 02.0);

3.2 Applicability of Methodology

The project satisfies all the applicability criteria of the methodology ACM0022 (Version 03.0), of which the detailed description is listed in the table below.

Table 5. Applicability of ACM0022

Application Criteria	Applicability
The methodology applies to project activities that install and operate new plants for the treatment of fresh waste through any combination of the following processes: (a) Composting process under aerobic conditions; (b) Anaerobic digestion with biogas recovery and flaring and/or its use; (c) Co-composting of wastewater in combination with solid waste; (d) Anaerobic co-treatment of wastewater in combination with solid waste; (e) Mechanical/thermal treatment process to produce refuse-derived fuel (RDF) or stabilized biomass (SB) that is produced within the project boundary and its use; (f) Gasification process to produce syngas and its use; (g) Incineration of fresh waste for the generation of thermal/electric energy. (h) The following conditions apply to all project activities using this methodology (i) The project plant only treats fresh waste/wastewater for which emission reductions are claimed, except for cases involving composting, co-composting and anaerobic digestion (j) Neither the fresh waste nor the products from the project plant are stored on-site under anaerobic conditions; (k) Any wastewater discharge resulting from the project activity is treated in accordance with applicable regulations; (l) The project activity does not reduce the amount of waste that would be recycled in the absence of the project activity. This shall be justified and documented in the clean development mechanism project design document (CDM-PDD); (m) When applicable regulations mandate any waste treatment process implemented under the project activity, the rate of compliance with such regulations for the treatment process is below 50 per cent;	Applicable. The project utilizes the composting process, biogas from anaerobic digestion of fresh waste for power generation and mechanical/thermal treatment process to produce refuse-derived fuel (RDF). Therefore, the project is applicable to (a), (b) and (e). For the condition (h): (i) The project is disposing fresh waste in the case of anaerobic digestion. Only fresh waste for which emission reductions are claimed. (j) The biogas residue produced in the project will be outward transported and will not be stored under anaerobic conditions. (k) The wastewater produced by the project will be treated aerobically by wastewater treatment plant nearby belongs ISTAC. (l) Prior to the implementation of the project, the fresh waste was disposed of at nearby landfill with a partial LFG capture system. The project will not reduce the amount of waste that would be recycled in the absence of the project activity. (m) Currently, there are no applicable laws or regulations in Turkey that require anaerobic digestion of fresh waste, and landfill is still the most common disposal method. (n) The fresh waste of the project does not involve hazardous waste.

Application Criteria	Applicability
(n) Hazardous wastes/wastewater are not eligible under this methodology.	
The methodology is only applicable if the baseline scenario is: (a) The disposal of the fresh waste in a SWDS with or without a partial LFG capture system (M2 or M3); (b) In the case of co-composting or co-treatment of wastewater in an anaerobic digester, the treatment of organic wastewater in either an existing or new anaerobic lagoon or sludge pit without methane recovery (W1 or W4); (c) In the case of electricity generation, the electricity is generated in an existing/new captive fossil fuel fired power-only plant, captive cogeneration plant and/or the grid (P2, P4 or P6); (d) In the case of heat generation, the heat is generated in an existing/new fossil fuel fired cogeneration plant, boiler or air heater (H2 or H4).	Applicable. The project utilizes the biogas from anaerobic digestion of fresh waste for power generation. The fresh waste was disposed at nearby landfill with a partial capture of the LFG and flaring of the captured LFG (M2); The electricity generated by the project is supplied to grid which is dominated by coal- fired power stations (P6). Therefore, the project is applicable to (a) and (c).
Specific applicability conditions for the different processes are provided in Table 2 of ACM0022 (Version 03.0).	The project utilizes the biogas from anaerobic digestion of fresh waste for power generation. Thus, the project is applicable to the waste treatment option “Anaerobic digestion” listed in Table 2 of ACM0022 (version 03.0).

In addition, the project meets the applicability conditions of the applied tools applied in the PD&MR as follows:

Table 6. Applicability of applied tools

Tool	Applicability	The Project
Project and leakage emissions from anaerobic digesters (version 02.0)	The following sources of project emissions are accounted for in this tool: (a) CO_2 emissions from consumption of electricity associated with the operation of the anaerobic digester; (b) CO_2 emissions from consumption of fossil fuels associated with the operation of the anaerobic digester; (c) CH_4 emissions from the digester (emissions during maintenance of the digester, physical leaks through the roof and side walls, and release through safety valves due to excess pressure in the digester); and (d) CH_4 emissions from flaring of biogas.	Applicable. The sources (a),(b), (c) and (d) of project emission are accounted for the project.
	The following sources of leakage emissions are accounted for in this tool: (a) CH_4 and N_2O emission from composting of digestate; (b) CH_4 emissions from the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic storage, such as in a stabilization pond.	Applicable The project involves anaerobic storage of digestate and plan composting; leakage emissions are going to be calculated when composting is installed.

Tool	Applicability	The Project
	Emission sources associated with N_2O emissions from physical leakages from the digester, transportation of feed material and digestate or any other on-site transportation, piped distribution of the biogas, aerobic treatment of liquid digestate and land application of the digestate are neglected because these are minor emission sources or because they are accounted in the methodologies referring to this tool.	Applicable. These are minor emission sources for the project.

Tool	Applicability	The Project
Emissions from solid waste disposal sites (version 08.0)	The tool can be used to determine emissions for the following types of applications: (a) Application A: The CDM project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g. "ACM0001: Flaring or use of landfill gas"). The methane is generated from waste disposed in the past, including prior to the start of the CDM project activity. In these cases, the tool is only applied for an ex ante estimation of emissions in the project design document (CDM-PDD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e.g. measuring the amount of methane captured from the SWDS); (b) Application B: The CDM project activity avoids or involves the disposal of waste at a SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW) is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions.	Applicable. The project is the condition of application B, that avoids or involves the disposal of waste at a SWDS.

Tool	Applicability	The Project
Tool to calculate the emission factor for an electricity system (version 07.0)	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects)	Applicable. The project utilizes the methane from anaerobic digestion of fresh waste for power generation supplied to grid that dominated by fossil-fuel plants. Therefore, the tool is applied to estimate the OM, BM and /or CM when calculating baseline emissions for the project. The Emission Factor is given from Turkish National Electricity Network calculated annually and for the year 2020 the combined emission factor is 0,5552 tCO_2/MWh[10].

[10]https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUESEmisyonFktr/Belgeler/Bform2020.pdf

Tool	Applicability	The Project
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub- options under the step 2 of the tool are available to the project participants, i.e. option II a and option II b. If option II a is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	In the host country as off-grid power generation is not significant. Therefore, emission factor for the project electricity system is calculated only for the grid power plants. Thus, this applicability criterion is satisfied.
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Not Applicable. The project electricity system is located in a Annex I country.
	Under this tool, the value applied to the CO2 emission factor of biofuels is zero.	The project does not involve biofuels.

Tool	Applicability	The Project
Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0)	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	Applicable. For the project, the electricity consumption is from the grid which the project is connected. Thus, Scenario A is applicable.

Tool	Applicability	The Project
	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	Applicable. For the project, the electricity is supplied to the grid. Thus, Scenario I is applicable.
	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO2 emissions.	The project does not involve captive renewable power generation technologies installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage.
Tool to determine the mass flow of a greenhouse gas in a gaseous stream (version 03.0)	Typical applications of this tool are methodologies where the flow and composition of residual or flared gases or exhaust gases are measured for the determination of baseline or project emissions.	Applicable. The volumetric flow of the gaseous stream and volumetric fraction of greenhouse gas CH_4 will be measured by the project.
	Methodologies where CO2 is the particular and only gas of interest should continue to adopt material balances as the means of flow determination and may not adopt this tool as material balances are the cost effective way of monitoring flow of CO2.	The biogas of the project produced by anaerobic digestion of fresh waste includes CH4, CO2, NH3 etc, CO2 is not the particular and only gas of interest by the project.

Tool	Applicability	The Project
	The underlying methodology should specify: (a) The gaseous stream the tool should be applied to; (b) For which greenhouse gases the mass flow should be determined; (c) In which time intervals the flow of the gaseous stream should be measured; and (d) Situations where the simplification offered for calculating the molecular mass of the gaseous stream (equations (3) or (17)) is not valid (such as the gaseous stream is predominantly composed of a gas other than N2).	Applicable. The methodology has specified (a) to (d).
Combined tool to identify the baseline scenario and demonstrate additionality (version 07.0)	The tool is applicable to all types of proposed project activities. However, in some cases, methodologies referring to this tool may require adjustments or additional explanations as per the guidance in the respective methodologies. This could include, inter alia, a listing of relevant alternative scenarios that should be considered in Step 1, any relevant types of barriers other than those presented in this tool and guidance on how common practice should be established.	Applicable. The project utilizes the methane from anaerobic digestion of fresh waste for power generation.
Project emissions from flaring" (version 03.0)	This tool provides procedures to calculate project emissions from flaring of a residual gas. The tool is applicable to enclosed or open flares and project participants should document in the CDM-PDD the type of flare used in the project activity.	Applicable. The project activity includes the enclosed flare and the type of which has been documented in the PD&MR.

Tool	Applicability	The Project
	This tool is applicable to the flaring of flammable greenhouse gases where: (a) Methane is the component with the highest concentration in the flammable residual gas; and (b) The source of the residual gas is coal mine methane or a gas from a biogenic source (e.g. biogas, landfill gas or wastewater treatment gas).	Applicable. The purpose of the project is to utilize biogas produced by anaerobic digestion, which consists mainly of methane, for electricity generation.
	The tool is not applicable to the use of auxiliary fuels and therefore the residual gas must have sufficient flammable gas present to sustain combustion. For the case of an enclosed flare, there shall be operating specifications provided by the manufacturer of the flare.	The project does not involve the use of auxiliary fuels. And the residual biogas has a methane content of over 50% and therefore has sufficient flammable gas present to sustain combustion.

3.3 Project Boundary

According to the methodology ACM0022 (version 03.0), the spatial extent of the project boundary is the SWDS where the waste is disposed of in the baseline, anaerobic lagoons or sludge pits treating organic wastewater in the baseline, and the site of the alternative waste treatment process(es). The boundary also includes on-site electricity and/or heat generation and use, on- site fuel use and the wastewater treatment plant used to treat the wastewater by-products of the alternative waste treatment process(es). The project boundary does not include facilities for waste collection and transport.

The spatial extent of the project boundary is the project site where the facility units of the project connected to the grid-system are located. The project boundary is the physical and geographical site of the methane recovery and renewable energy production, respectively. Both the physical and geographical project boundaries is schematically illustrated in the figure below.

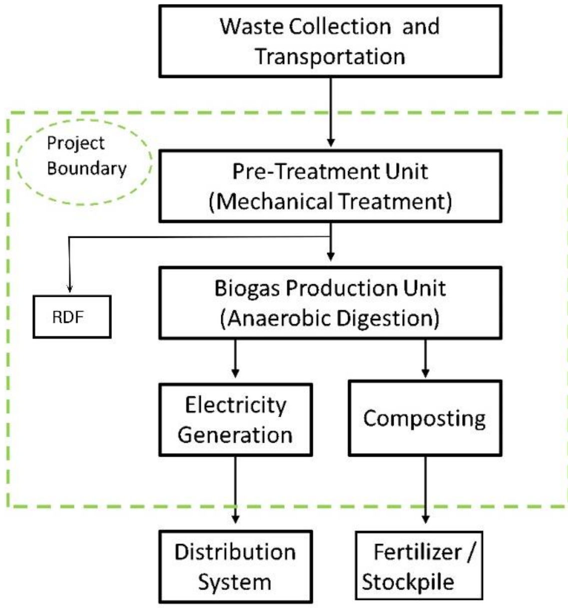

Figure 10. Project Boundary

Table 7. Emission sources included in or excluded from the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from heat generation	CO_2	No	Excluded for simplification. This is conservative.
		CH_4	No	
		N_2O	No	
		Other	No	
	Emissions at the SWDS	CO_2	No	CO2 emissions from the decomposition of fresh waste are not accounted for.
		CH_4	Yes	Major emission source.
		N_2O	No	N2O emissions are small compared to CH4 emissions from biogas. Exclusion of this gas is conservative.
		Other	No	-
	Emissions from anaerobic lagoons or sludge pits	CO_2	No	The project is not involved baseline emissions from anaerobic lagoons or sludge pits.
		CH_4	No	

Source		Gas	Included?	Justification/Explanation
		N_2O	No	
		Other	No	
	Emissions from electricity generation	CO_2	Yes	Major source, electricity generation is included in the project activity and is sent to the grid or displaces fossil fuel fired electricity generation in the baseline.
		CH_4	No	Excluded for simplification. This is conservative.
		N_2O	No	Excluded for simplification. This is conservative.
		Other	No	-
	Emissions from use of natural gas	CO_2	No	The project is not involved baseline emissions from use of natural gas.
		CH_4	No	
		N_2O	No	
		Other	No	
Project	Emissions from on-site fossil fuel consumption due to the project activity other than for electricity generation	CO_2	Yes	Included. This system is used when the grid cannot be utilized in case of power failures.
		CH_4	No	Excluded for simplification. This emission source is assumed to be very small
		N_2O	No	Excluded for simplification. This emission source is assumed to be very small
		Other	No	-
	Emissions from on-site electricity use	CO_2	Yes	May be an important emission source.
		CH_4	No	Excluded for simplification. This emission source is assumed to be very small.

Source		Gas	Included?	Justification/Explanation
		N_2O	No	Excluded for simplification. This emission source is assumed to be very small.
		Other	No	-
	Emissions from the waste treatment processes	CO_2	No	The project is not involved incineration, gasification or combustion of fossil based waste.
		CH_4	Yes	CH4 leakage from the anaerobic digester and incomplete combustion in the flaring process are potential source of project emissions.
		N_2O	No	The project is not involved incineration, gasification or combustion of fossil based waste.
		Other	No	-
	Emissions from wastewater treatment	CO_2	No	CO2 emissions from the decomposition of fresh waste are not accounted.
		CH_4	No	The project does not involve anaerobic treatment of wastewater.
		N_2O	No	Excluded for simplification. This emission source is assumed to be very small.
		Other	No	-

3.4 Baseline Scenario

According to the methodology ACM0022 (Version 03.0), using “TOOL02: Combined tool to identify the baseline scenario and demonstrate additionality” (version 07.0) to identify the baseline scenario and demonstrate additionality.

According to tool 02, the project participants shall apply the following four Steps:

STEP 0. Demonstration that a proposed project activity is the first-of-its-kind;

STEP 1. Identification of alternative scenarios;

STEP 2. Barrier analysis;

STEP 3. Investment analysis (if applicable); STEP 4. Common practice analysis.

Step 0: Demonstration whether the proposed project activity is the first-of-its-kind

- The project is not the first-of-its-kind project.

Step 1: Identification of alternative scenarios

This step serves to identify all alternative scenarios to the project activity(s) which can be the baseline scenario.

Step 1a: Define alternative scenarios to the proposed VCS project activity

Since the project only involves the treatment of fresh waste (anaerobic digestion) and the generation of electricity, the baseline scenario will be identified based on these two parts.

In identifying baseline alternatives for the treatment of the fresh waste, the following alternatives or combinations of these alternatives shall, inter alia, be considered:

Alternative scenarios provided by methodology ACM0022 (Version 03.0)		Realistic and credible?	Alternatives to Electricity Generation Component
M1	The project activity without being registered as a VCS project activity (i.e. any (combination) of the waste treatment options listed in Table 2 of methodology ACM0022).	Yes	Anaerobic digestion treatment method adopted in this project is an encouraging policy in Turkey, therefore, M1 is a feasible baseline scenario.

M2	Disposal of the fresh waste in a SWDS with a partial capture of the LFG and flaring of the captured LFG.	Yes	At present, there are no laws and regulations in our country that require landfills to recycle landfill gas. It is demonstrated that sufficient landfill capacity would be available to dispose waste at a SWDS with a comparable annual waste acceptance rate and with the same operating lifetime as the project activity. And there may be monitored data for LFG recovered by LFG capture system and flaring or power generation system, however, since LFG recovery and power generation system is owned by another company, the data is out of reach. Furthermore, it is hard to capture and monitor all LFG generated by the SWDS. Hence, to be conservative, the SWDS is with a partial capture of the LFG and flaring of the captured LFG. To be conservative, M2 is a realistic and credible alternative scenario.
M3	Disposal of the fresh waste in a SWDS without a LFG capture system.	Yes	M3 is a realistic and credible scenario. The alternative doesn’t require any investment. There are no mandatory regulatory or contractual requirements that mandate to undertake the alternative. Therefore, M3 is considered for further assessment.

M4	Part of the fresh fraction of the solid waste is recycled and not disposed in the SWDS.	Yes	If part of the fresh portion of the solid waste is in a condition that can be recycled, it is recycled and not disposed of in the SWDS.
M5	Part of the fresh fraction of the solid waste is treated aerobically and not disposed in the SWDS.	No	According to project, there is no such facility at the location of this project, therefore, M5 is not a realistic and credible alternative scenario.
M6	Part of the organic fraction of the solid waste is incinerated and not disposed in the SWDS.	No	According to project, there is no such facility at the location of this project, therefore, M6 is not a realistic and credible alternative scenario.
M7	Part of the organic fraction of the solid waste is gasified and not disposed in the SWDS.	No	According to project, there is no such facility at the location of this project, therefore, M7 is not a realistic and credible alternative scenario.
M8	Part of the organic fraction of the solid waste is treated in an anaerobic digester and not disposed in the SWDS.	No	According to project, before the implementation of the project, there is no such facility at the location of this project, therefore, M8 is not a realistic and credible alternative scenario.
M9	Part of the organic fraction of the solid waste is mechanically or thermally treated to produce RDF/SB and not disposed in the SWDS.	Yes	According to project, before the implementation of the project, there was such facility at the location of this project, therefore, M9 is a realistic and credible alternative scenario.

Electricity generation is an aspect of the project activity, then alternative scenarios for the generation of electricity shall also be identified. Alternative(s) shall include, inter alia:

Alternative scenarios provided by methodology ACM0022 (Version 03.0)		Realistic and credible?	Justification/explanation
P1	Electricity generated as an output of one of the waste treatment options listed in Table 2 of methodology ACM0022 (Version 03.0), not undertaken as a VCS project activity.	Yes	The methane from anaerobic digestion will be utilized for generation, which is an encouraging policy in Turkey, therefore, P1 is a feasible baseline scenario.
P2	Use of an existing or construction of a new on- site or off-site fossil fuel fired cogeneration plant.	Yes	The project involved power generation, and cogeneration plant. Thus, P2 is a realistic and credible alternative scenario.
P3	Existing or new construction of an on-site or off-site renewable based cogeneration plant.	Yes	The project involved power generation, and a cogeneration plant. Thus, P3 is a realistic and credible alternative scenario.
P4	Existing or new construction of an on-site or off-site fossil fuel fired electricity plant.	No	Construction of a power plant with the same capacity as the Project using coal, diesel oil or natural gas as fuel would be prohibited. Thus, P4 is not a realistic and credible alternative scenario.

P5	Existing or new construction of an on-site or off-site renewable based electricity plant.	No	Wind, solar photovoltaic, hydro, biomass and geothermal energy are feasible renewable energy sources that can be applied. The wind resources of İstanbul Province is in the provincial capital city area, which is not suitable for the construction of wind farms. The solar energy of İstanbul Province is in the provincial capital city area, which is not suitable for the construction of and solar plants. The facility is located in the provincial capital city area, which is not suitable for the construction of hydropower stations. The project is located in the provincial capital city area, the project site has no adequate biomass materials supplied for a biomass power. Thus, P5 is not a realistic and credible alternative scenario.
P6	Electricity generation in existing and/or new grid- connected electricity plants.	Yes	P6 is the actual situation before the start of the project activities and meets the requirements of national regulations, so it is a feasible alternative. Electricity generated in new grid-connected electricity plant.

Outcome of Step 1a: List of plausible alternative scenarios to the project activity

According to the analysis above, M1 and M2 for the treatment of the fresh waste, P1 and P6 for electricity generation are feasible baseline scenarios.

Step 1b: Consistency with mandatory applicable laws and regulations

M1 and M8 for the treatment of the fresh waste, P1 and P6 for electricity generation are all consistent with mandatory applicable laws and regulations.

Scenario ID	Treatment of the fresh waste	Electricity generation
Combined alternative scenario 1 (M1+P1): This is the project activity not implemented as a VCS project. This scenario faces investment barriers, which will be discussed in section 3.5.	M1	P1
Combined alternative scenario 2 (M1+P6): This is a conflicted scenario for use of methane for electricity production conflicted with equivalent quantity of electricity supply from existing and/or new grid-connected electricity plants.	M1	P6
Combined alternative scenario 3 (M2+P1): This is a conflicted scenario, for the disposal of the fresh waste in a SWDS with a partial capture of the LFG and flaring of the captured LFG is conflicted with the fresh waste will be treated by way of anaerobic digestion, the two scenarios will not be existed at the same time.	M8	P1
Combined alternative scenario 4 (M2+P6): This is the continuation of business as usual. It remains as a possible baseline scenario alternative.	M8	P6

Outcome of Step 1b: List of alternative scenarios to the project activity that follow mandatory legislation and regulations considering the enforcement in the region or country and Board decisions on national and/or sectoral policies and regulations.

As above, the baseline alternative options that are technically feasible and comply with all legal and regulatory requirements are combined alternative Scenario 1 (M1+P1) and Scenario 4 (M8+P6).

3.5 Additionality

As per 25th and 26th paragraphs of ACM0022 (V 03.0);

"25. Project activities to implement a Greenfield composting facility to treat MSW are deemed automatically additional, if any of conditions below is fulfilled:

(a) The host country is a LDC; or

(b) If MSW collection coverage is below 50 per cent for the applicable geographical region; or

(c) If MSW collection coverage is 50–80 per cent for the applicable geographical region[11] and if the waste received by the project composting facility does not have formal (i.e. excluding recycling by the informal sector) segregation of wet and dry waste; or

(d) If no gate/tipping fees are received by the project composting facility or any upstream waste collection or segregation facility which is contractually bound to provide waste to the project composting facility[12]; or

(e) If less than two per cent of the collected MSW of the municipality (or the region from where the MSW treated by the project activity is sourced)[13] are treated by composting in the year before the PDD is published for global stakeholder consultation.

26. If any of the above conditions applies, the baseline scenario is assumed to be disposal of the fresh waste in the nearest SWDS (where the LFG is released, or captured and destructed through flaring to comply with regulations or contractual requirements, to address safety and odour concerns, or for other reasons). If condition (d) or (e) applies, it shall also be demonstrated that 80 per cent of the collected MSW of the municipality (or the region from where the MSW treated by the project activity is sourced) are treated in SWDS or stockpiles or are dumped in the year before the PDD is published for global stakeholder consultation.

The project is a Greenfield project implementing composting facility as an alternative municipal solid waste treatment.

As required in the 25th paragraph of the methodology condition (d) and (e) apply. Therefore, the baseline scenario is assumed to be disposal of the fresh waste in the nearest SWDS.

[11] May be certified by a municipal authority or another MSW treatment facility in the municipality.

[12] May be certified by a municipal authority or another MSW treatment facility in the municipality.

[13] The applicable geographical region may be the host country, the region or the municipality where the project activity is located.

Moreover, more than the 80% of the collected MSW of the are treated in SWDS or stockpiles or are dumped in the year before the PDD is published for global stakeholder consultation.

Hence, the proposed project meets the criteria for deemed additionality.

3.6 Methodology Deviations

No methodology deviation is applied in the project.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

It is stated in the methodology ACM0022 (V 03.0) baseline emissions are calculated according to equation (1). Based on methodology the baseline scenario for the project is:

$$BE_y = \sum_t \left(BE_{CH4,t,y} + BE_{WW,t,y} + BE_{EN,t,y} + BE_{NG,t,y}\right) \times (1 - RATE_{compliance,t}) \qquad \text{Equation (1)}$$

$$DF_{RATE,t,y} = 1 - RATE_{compliance,t,y}, \quad if\ RATE_{compliance,t,y} < 0.5$$

$$DF_{RATE,t,y} = 0, \quad if\ RATE_{compliance,t,y} \geq 0.5 \qquad \text{Equation (2)}$$

Where:

BE_y	=	Baseline emissions in year *y* (t CO_2e)
$BE_{CH4,t,y}$	=	Baseline emissions of methane from the SWDS in year *y* (t CO_2e)
$BE_{WW,t,y}$	=	Baseline methane emissions from anaerobic treatment of the wastewater in open anaerobic lagoons or of sludge in sludge pits in the absence of the project activity in year *y* (t CO_2e)
$BE_{EN,t,y}$	=	Baseline emissions associated with energy generation in year *y* (t CO_2)
$BE_{NG,t,y}$	=	Baseline emissions associated with natural gas use in year *y* (t CO_2)

$RATE_{compliance,t}$	=	Discount factor to account for the rate of compliance of a regulatory requirement that mandates the use of alternative waste treatment process t [14]
t	=	Type of alternative waste treatment process

In the case that legislative or regulatory requirements mandate the use of any (combination) of the waste treatment options t that are being implemented in the project activity, then the rate of compliance with these requirements in the host country shall be monitored ($RATE_{compliance,t,y}$). This rate is then used to adjust the baseline emissions calculation, according to equation 1. It shall be described and justified in the PDD which baseline emission sources apply for each waste treatment option t implemented under the project activity.

In the project context, there are no regulations that mandate the use of the project activity treatment option.

Therefore $RATE_{compliance,t,y} = 0$.

So that;

$DF_{RATE,t,y}$ = 1– 0 = 1 Equation (3)

Additionally, project activity does not include treatment of the wastewater and natural gas replacement. Therefore;

$BE_{WW,y} = 0$

$BE_{NG,y} = 0$

Thus, baseline emissions are;

- Emissions of methane from the SWDS,
- Emissions associated with energy generation.

4.1.1 Baseline emissions of methane from the SWDS (BECH4,t,y)

Baseline emissions of methane from the SWDS are determined using the methodological tool “Emissions from solid waste disposal sites”. Additionally, “emissions are calculated using Application B in the tool, meaning that only waste avoided from the disposal after the start

[14] Determined once for each crediting period, based on the most recent data available at the time of submission of the CDM-PDD to the DOE for validation.

of the first crediting period shall be considered in the tool". The tool can be applied for both ex ante and ex post estimation of emissions.

The amount of methane generated from disposal of waste at the SWDS is calculated based on a first order decay (FOD) model in line with the methodological tool "Emissions from solid waste disposal sites", V.08.0.

$$BE_{CH4,SWDS,y} = \varphi y\,(1 - fy) * GWP\ CH4 * (1 - OX) * 16/12 * F * DOCf,y * MCFy * \sum\sum Wj,x * DOCj * exp(-kj\,(y - x)) * (1 - exp(-kj))$$

Equation (4)

$BE_{CH4,SWDS,y}$	=	Baseline methane emissions occurring in year y generated from waste disposal at the solid waste disposal site (SWDS) during a period ending in year y (tCO_2e/y)
φ	=	Model correction factor to account for model uncertainties (default value of 0.85), Option 1 in the Tool has been selected, value as per Table 3 of the Tool (Application B and humid wet conditions).
f	=	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y.
GWP_{CH4}	=	Global warming potential of CH_4 (tCO_2e/tCH_4)
OX	=	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
F	=	Fraction of methane in the SWDS gas (volume fraction) (0.5)
$DOCf,y$	=	Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year y (weight fraction). Default value of 0.5 used as per page 65 of the Tool.
$MCFy$	=	Methane correction factor for year y (1.0)
Wj,x	=	Amount of solid waste type j disposed or prevented from disposal in the SWDS in the year x (t)
DOC	=	Fraction of degradable organic carbon (by weight fraction) in the waste type j
kj	=	Decay rate for the waste type j (1/yr)
j	=	Type of residual waste or types of waste in the MSW
x	=	Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period (x=1) to year (x = y)
y	=	Year for which methane emissions are calculated (considering a consecutive period of 12 months)

Table 8. Determining the parameters required to apply the FOD model

Parameter	Value	Determination Application B
φy	0.85	Default value as per "Emission from solid waste disposal sites" Ver.08.0
OX	0.1	Default value as per "Emission from solid waste disposal sites" Ver.08.0
GWPCH4	28 tCO2e/tCH4	https://www.ipcc.ch/report/ar5/syr/
F	0.5	Default value as per "Emission from solid waste disposal sites" Ver.08.0
$DOC_{f,y}$	0.5	Default value as per "Emission from solid waste disposal sites" Ver.08.0
MCFy	1	Default values (based on SWDS type)
fy	0.0	Default value as per "Emission from solid waste disposal sites" Ver.08.0

$BE_{CH4,SWDS,y}$ = 273,073 tCO_2e (annual average)

4.1.2 Baseline emissions from generation of energy

This procedure is distinguished depending on whether the baseline is the separate generation of electricity and heat or the combined generation of heat and electricity by cogeneration.

The project is using methane from fresh waste by anaerobic digestion treatment for electricity generation. Thus, baseline emissions from separate generation of electricity ($BE_{EC,y}$)is considered.

4.1.2.1 Separate generation of electricity and heat

$$BE_{EN,y} = BE_{EC,y} + BE_{HG,y} \qquad \text{Equation (5)}$$

Where:

$BE_{EN,y}$	=	Baseline emissions associated with energy generation in year *y* (t CO_2)
$BE_{EC,y}$	=	Baseline emissions associated with electricity generation in year *y* (t CO_2)
$BE_{HG,y}$	=	Baseline emissions associated with heat generation in year *y* (t CO_2)

As per mentioned above, there is no baseline emissions associated with heat generation thus $BE_{HG,y}$=0 and consequently $BE_{EN,y}$= $BE_{EC,y}$.

Baseline emissions associated with electricity generation ($\boldsymbol{BE_{EC,y}}$)

The baseline emissions associated with electricity generation in year y ($BE_{EC,y}$) shall be calculated using the "Methodological tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation". When applying the tool:

(a) The electricity sources k in the tool correspond to the sources of electricity generated identified in the selection of the most plausible baseline scenario; and

(b) $EC_{BL,k,y}$ in the tool is equivalent to the net amount of electricity generated using LFG in year y ($EG_{PJ,y}$).

Taking into account the approach provided by the tool, baseline emissions are then calculated using the generic approach based on the quantity of electricity dispatched into the National Grid, an emission factor for electricity generation and a factor to account for transmission losses, as follows:

$$BE_{EC,y} = \sum EC_{BL,k,y} * EF_{EL,k,y} * (1 + TDL_{k,y}) \quad \text{Equation (6)}$$

Where;

$EC_{BL,k,y}$	=	Net amount of electricity generated using LFG in year y (MWh/yr)
$EF_{EL,k,y}$	=	Emission factor for electricity generation for source k in year y (tCO_2/MWh)
$TDL_{k,y}$	=	Average technical transmission and distribution losses for providing electricity to source k in year y
k	=	Sources of electricity generated in the baseline

Emission Factor calculation

The Emission Factor is calculated as the Combined Margin (CM), comprised by two components: the Built Margin (BM) and the Operation Margin (OM). The BM evaluates the contribution of the power plants which would have been built if the project plant would not have been implemented. The OM evaluates the contribution of the power plants which would have been dispatched in the absence of the project activity. TOOL07 presents the following steps to calculate the Emission Factor:

- STEP 1: Identify the relevant electricity systems:

The sources of power in the Turkish electric system were taken from the Turkish Electricity transmission corporation. https://www.teias.gov.tr/en-US

- STEP 2: Choose whether to include off-grid power plants in the project electricity system (optional)

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

Option I of the tool is chosen, which is to include only grid power plants in the calculation.

- STEP 3 - Select a method to determine the operating margin (OM). The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods:

(a) Simple OM, or

(b) Simple adjusted OM, or

(c) Dispatch data analysis OM, or

(d) Average OM. The simple operating margin can only be used where low-cost/must-run resources15 constitute less than 50% of total grid generation in:

1) average of 5 most recent years, or

2) based on long-term normality's for hydroelectricity production. Figure 9 shows the share of the electricity production in the turkey interconnected system. The results show the applicability of the simple operating margin.

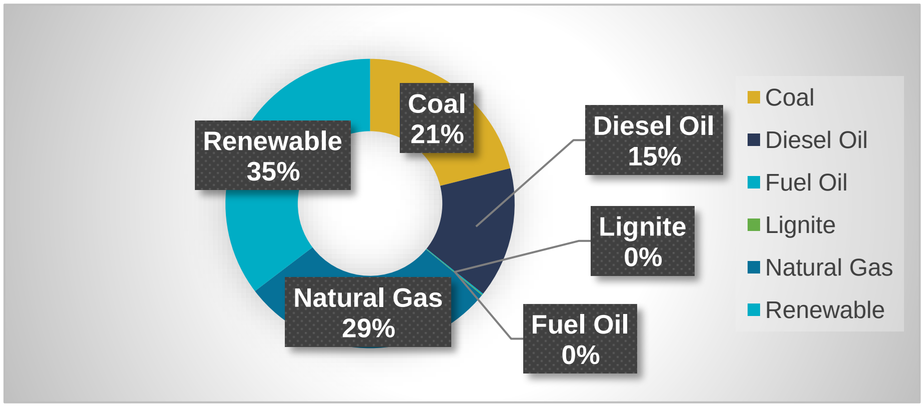

Figure 11. Electricity generation in Turkish interconnected system by source[15]

[15] http://www.teias.gov.tr/sites/default/files/2019-03/56%2893-2017%29.xls

- STEP 4 - Calculate the operating margin emission factor according to the selected method

Simple OM

The simple OM emission factor is calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (tCO2/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units.

The simple OM may be calculated by one of the following two options:

(a) Option A: Based on the net electricity generation and a CO_2 emission factor of each power unit; or

(b) Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system. Option B can only be used if:

a. The necessary data for Option A is not available; and

b. Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and

c. Off-grid power plants are not included in the calculation (i.e. if Option I has been chosen in Step 2)

Under this option, the simple OM emission factor is calculated based on the net electricity generation of each power unit and an emission factor for each power unit, as follows:

$$EF_{grid,OM\ simple,y} = \frac{\sum_m EG_{m,y} * EF_{EL,m,y}}{\sum_m EG_{m,y}}$$

Where:

$EF_{grid,OM\ simple,y}$	=	Simple adjusted operating margin CO_2 emission factor in year y (tCO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO2 emission factor of power unit m in year y (tCO_2/MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
y	=	The relevant year as per the data vintage chosen in Step 3

- STEP 5 - Calculate the build margin (BM) emission factor The sample group of power units m used to calculate the build margin was determined following the procedure provided by the tool and BM emission factor shall be calculated based on the equation below:

$$EF_{grid,BM\,simple,y} = \frac{\sum_m EG_{m,y} * EF_{EL,m,y}}{\sum_m EG_{m,y}}$$

Where:

$EF_{grid,BM,y}$=	=	Build margin CO_2 emission factor in year y (tCO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (tCO2/MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
y	=	The relevant year as per the data vintage chosen in Step 3

- STEP 6 – Calculate the combined margin (CM) emissions factor. The calculation of the combined margin (CM) emission factor is based on one of the following methods:

  (a) Weighted average CM; or

  (b) Simplified CM.

Since power grid is not located in LDC/SIDs/URC and the weighted average CM method (option A) is the preferred option, this method was considered. The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} * w_{OM} + EF_{grid,BM,y} * w_{BM}$$

Where:

$EF_{grid,OM,y}$	=	Operating margin CO_2 emission factor in year y (tCO_2/MWh)
w_{OM}	=	Weighting of operating margin emissions factor (%)
$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year y (tCO_2/MWh)
w_{BM}	=	Weighting of build margin emissions factor (%)

As per the applicable “Tool to calculate the emission factor for an electricity system”, version 07.0, the default weights for the operating margin and build margin emission factors for the proposed project activity is defined as:

w_{OM} = 0.50

w_{BM} = 0.50

The Emission Factor is given from Turkish National Electricity Network calculated annually and for the year 2020 the combined emission factor is 0,5552 tCO2/MWh[16].

$\boxed{BE_{electricity,y} = 62{,}579.956\ tCO_2e\ \text{(annual average)}}$

4.2 Project Emissions

The project emissions in year y are calculated for each alternative waste treatment option implemented in the project activity as follows:

$$PE_y = PE_{COMP,y} + PE_{AD,y} + PE_{GAS,y} + PE_{RDF\_SB,y} + PE_{INC,y} \qquad \text{Equation (7)}$$

Where:

PE_y	=	Project emissions in year *y* (t CO_2e)
$PE_{COMP,y}$	=	Project emissions from composting or co-composting in year *y* (t CO_2e)
$PE_{AD,y}$	=	Project emissions from anaerobic digestion and biogas combustion in year *y* (t CO_2e)
$PE_{GAS,y}$	=	Project emissions from gasification in year *y* (t CO_2e)
$PE_{RDF\_SB,y}$	=	Project emissions associated with RDF/SB in year *y* (t CO_2e)
$PE_{INC,y}$	=	Project emissions from incineration in year *y* (t CO_2e)

Project activity does not include gasification and incineration.

So that;

$$PE_y = PE_{AD,y} + PE_{RDF\_SB,y} + PE_{COMP,y} \qquad \text{Equation (8)}$$

4.2.1 Project emissions from anaerobic digestion ($PE_{AD,y}$)

$PE_{AD,y}$ is calculated according to the methodological tool “Project and leakage emissions from anaerobic digesters” (V02.0).

$$PE_{AD,y} = PE_{EC,y} + PE_{FC,y} + PE_{CH4,y} + PE_{flare,y} \qquad \text{Equation (9)}$$

[16]https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUESEmisyonFktr/Belgeler/Bform2020.pdf

Where:

$PE_{AD,y}$	=	Project emissions associated with the anaerobic digester in year y (t CO_2e)
$PE_{EC,y}$	=	Project emissions from electricity consumption associated with the anaerobic digester in year y (t CO_2e)
$PE_{FC,y}$	=	Project emissions from fossil fuel consumption associated with the anaerobic digester in year y (t CO_2e)
$PE_{CH4,y}$	=	Project emissions of methane from the anaerobic digester in year y (t CO_2e)
$PE_{flare,y}$	=	Project emissions from flaring of biogas in year y (t CO_2e)

4.2.1.1 Project emissions from electricity consumption associated with the anaerobic digester ($PE_{EC,y}$)

Project emissions from electricity consumption will be calculated as a bundle for the all units included in the project activity.

4.2.1.2 Project emissions from fossil fuel consumption associated with the anaerobic digester ($PE_{FC,y}$)

According to regulation there is to have a captive power plant or urgency diesel generator in case of shortage on methane for electricity production in case the electricity needs to be maintained and the engines don't provide it temporarily.

CO2 emissions from fossil fuel combustion in process j are calculated based on the quantity of fuels combusted and the CO2 emission coefficient of those fuels, as follows:

$$PE_{FC,j,y} = \sum_j FC_{PJ,j,y} * COEF_{i,y} \qquad \text{Equation (9)}$$

Where:

$PE_{FC,j,y}$	=	Are the CO2 emissions from fossil fuel combustion in process j during the year y (tCO2/yr)
$FC_{PJ,j,y}$	=	Is the quantity of fuel type i combusted in process j during the year y (mass or volume unit/yr)

$COEF_{i,y}$ = Is the CO2 emission coefficient of fuel type i in year y (tCO2/mass or volume unit)

i = Are the fuel types combusted in process j during the year y

$$COEF = 0{,}00268 \frac{\text{tCO2}}{\text{L}} fuel$$

As per tool 3, in order to determine the parameter CO_{EF}, option A is the preferred approach as the necessary data is available.

4.2.1.3 Project emissions from flaring ($PE_{flare,y}$)

Methodological TOOL06 show the calculation procedure to determine the project emissions form flaring the residual gas PE, flare based on the flare efficiency (flare) and the mass flow of methane to the flare (FCH4,RG,m). The flare efficiency is determined based on monitored data or default values.

The methodological TOOL shows a procedure form three steps to calculate those emissions. The tool to be used to calculates the project emissions from flaring, if working hours wouldn't have been 0.

(a) STEP 1: Determination of the methane mass flow of the residual gas;

(b) STEP 2: Determination of the flare efficiency;

(c) STEP 3: Calculation of project emissions from flaring.

a. STEP 1: Determination of the methane mass flow of the residual gas

The residual gas is monitored continuously in normal conditions (Nm3/h) every 10 min for 24/7. The mass flow of methane in the residual gas in (kg/m) is calculated.

$F_{CH4,RG,t} = V_{t,db} * v_{CH4,wb,t} * \rho_{CH4,n}$	Equation (10)

Where:

$\mathbf{F_{CH4,t}}$ = Mass flow of greenhouse gas (*CH4)* in the gaseous stream in time interval *t* (kg gas/h)

$\mathbf{V_{t,wb}}$ = Volumetric flow of the gaseous stream in time interval *t* on a dry basis (m³ dry gas/h)

$\mathbf{v_{CH4,t,wb}}$ = Volumetric fraction of greenhouse gas *CH4* in the gaseous stream in a time interval *t* on a dry basis (m³ gas *i*/m³ dry gas)

$\rho_{CH4,n}$	Density of greenhouse gas CH4 in the gaseous stream at normal conditions t (kg gas i/m³ gas i)

b. **STEP 2: Determination of the flare efficiency;**

- ✓ Enclosed flare

In this case the project implemented an enclosed flare, there are two options to determine the flare efficiency for minute m ($\eta_{flare,m}$)

(a) Option A: Apply a default value for flare efficiency;

(b) Option B: Measure the flare efficiency.

Option A is chosen because:

(a) The temperature of the flare (TEG,m) and the flow rate of the residual gas to the flare (FRG,m) is within the ARİŞ Enerji's operating specification for the flare ($SPEC_{flare}$) in the minute m;

(b) The flame is detected in the minute m (Flame).

The set of thermocouples, measure continuously temperature and allow the detection of the presence and absence of flame, which can be cross check with the flare working hours and amount of residual gas to flare. T (0) flare is the parameter can be read form the compiled store data from the control system.

Since the enclose flare could be define as low height flare the efficiency is adjusted 10 percentile points less form the default value:

$\eta_{flare,m}$	80%

c. STEP 3: Calculation of project emissions from flaring.

$$PE_{flare,y} = GWP_{CH4} * \sum_{m=1}^{525600} F_{CH4,RG,m} * (1 - \eta_{flare,m}) * 10^{-3}$$	Equation (11)

Where:

$PE_{flare,y}$	=	Project emissions from flaring of the residual gas in year y (tCO2e)
GWP_{CH4}	=	Global warming potential of methane valid for the commitment period (tCO2e/tCH4)

$FCH4,RG,m$ = Mass flow of methane in the residual gas in the minute m (kg)

$\eta flare,m$ = Flare efficiency in the minute m

Parameter flare working hours and hour meter flare can be read directly for crosschecking.

Therefore:

Flare working hours (h)	0
$PE_{flare,y}$ ((tCO2e))	0

$PE_{FC,j,y}$ is determined by "TOOL03 Methodological tool: Tool to calculate project or leakage CO2 emissions from fossil fuel combustion. Version 03.0". They are the CO_2 emissions from fossil fuel combustion in process j during the year y (tCO2/yr).

According to regulation there is to have a captive power plant or urgency diesel generator in case of shortage on methane or a break out. this only implement diesel and has not been used once, please refer to monitoring reports. Therefore, CO2 emissions from fossil fuel combustion in process is taken as zero.

$$\boxed{PE_{FC,j,y} = 0}$$

4.2.1.4 Project emissions of methane from the anaerobic digester (PECH4,y)

Project emissions of methane from the anaerobic digester include emissions during maintenance of the digester, physical leaks through the roof and side walls, and release through safety valves due to excess pressure in the digester. These emissions are calculated using a default emission factor ($EF_{CH4,default}$), as follows:

$$PE_{CH4,y}=Q_{CH4,y}\times EF_{CH4,default}\times GWP_{CH4} \qquad \text{Equation (12)}$$

Where:

$PE_{CH4,y}$ = Project emissions of methane from the anaerobic digester in year y (t CO_2e)

$Q_{CH4,y}$ = Quantity of methane produced in the anaerobic digester in year y (t CH_4)

$EF_{CH4,default}$ = Default emission factor for the fraction of CH_4 produced that leaks from the anaerobic digester (fraction)

GWP_{CH4} = Global warming potential of CH_4 ($t\ CO_2$ / $t\ CH_4$)

$\boxed{PE_{CH4} = 12{,}845.469\ tCO_2e\ \text{(annual average)}}$

4.2.2 Project emissions associated with mechanical or thermal production of RDF/SB ($PE_{RDF\_SB,y}$)

Project emissions associated with RDF/SB[17] comprise both the emissions from the mechanical/thermal production process (e.g. electricity, fossil fuel consumption and wastewater treatment, if relevant) as well as the combustion of RDF/SB. Project emissions are determined as follows:

$$PE_{RDF\_SB,y} = PE_{COM,RDF\_SB,y} + PE_{EC,RDF\_SB,y} + PE_{FC,RDF\_SB,y} + PE_{ww,RDF\_SB,y}$$

Equation (13)

Where:

$PE_{RDF\_SB,y}$	=	Project emissions associated with RDF/SB in year *y* ($t\ CO_2e$)
$PE_{COM,RDF\_SB,y}$	=	Project emissions from combustion of fossil waste associated with combustion of RDF/SB within the project boundary in year *y* ($t\ CO_2$)
$PE_{EC,RDF\_SB,y}$	=	Project emissions from electricity consumption associated with RDF/SB (production and on-site combustion) in year *y* ($t\ CO_2e$)
$PE_{FC,RDF\_SB,y}$	=	Project emissions from fossil fuel consumption associated with RDF/SB (production and on-site combustion) in year *y* ($t\ CO_2e$)
$PE_{ww,RDF\_SB,y}$	=	Project emissions from the wastewater treatment associated with RDF/SB (production and on-site combustion) in year *y* ($t\ CH_4$)

4.2.3 Project emissions from composting

$$PE_{COMP,y} = PE_{EC,y} + PE_{FC,y} + PE_{CH4,y} + PE_{N20y} + PE_{RO,y}$$

Where:

$PE_{COMP,y}$	=	Project emissions associated with composting in year y ($t\ CO_2e/yr$)
$PE_{EC,y}$	=	Project emissions from electricity consumption associated with composting in year y ($t\ CO_2/yr$)

[17] RDF system is being planned to get active in the first quarter of the 2023.

$PE_{FC,y}$ = Project emissions from fossil fuel consumption associated with composting in year y (t CO_2/yr)

$PE_{CH4,y}$ = Project emissions of methane from the composting process in year y (t CO_2e/yr)

$PE_{N20,y}$ = Project emissions of nitrous oxide from the composting process in year y (t CO_2e/yr)

$PE_{RO,y}$ = Project emissions of methane from run-off wastewater associated with co-composting in year y (t CO_2e/yr)

Project emissions from electricity consumption will be calculated as a bundle for the all units included in the project activity. Due to composting facility is being considered for June 2023, calculations are going to be included after composting units commissioned.

Project emissions from electricity consumption ($PE_{EC,y}$)

As it was stated above Project emissions from electricity consumption is calculated as a bundle for the all units included in the project activity.

ACM0022 (V03.0) refers the methodological tool "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (V 03.0)" to calculate emissions from consumption of electricity due to the project activity and this tool directs to the methodological tool "Tool to calculate the emission factor for an electricity system (V 07.0)".

"Tool to calculate the emission factor for an electricity system (V 07.0)" is applied in the above section to calculate the combined margin emission factor of the applicable electricity system for the calculations of baseline emissions. Since the electricity to be consumed is being supplied by the same grid that the generated electricity fed into, same e combined margin emission factor is being used to calculate the project emissions from electricity consumption.

Thus:

$$PE_{EC,} = \sum EC_{PJ,j,y} \times EF_{EF,j,y} \times (1 + TDL_{j,y}) \qquad \text{Equation (14)}$$

Where:

$PE_{EC,y}$ = Project emissions from electricity consumption in year y (tCO_2/yr)

$ECPJ_{,j,y}$ = Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr)

$EFEF_{,j,y}$ = Emission factor for electricity generation for source j in year y (tCO_2/MWh)

$TDL_{j,y}$ = Average technical transmission and distribution losses for providing electricity to source j in year y

According to the TEIAS data[18] the transmission loss rate is 1.7% and distribution loss rate is 10.2% for the year 2018. Thus;

$TDLj,$ = 11.9%

$\boxed{PE_{EC,y} = 28.172\ tCO_2e\ \text{(annual average)}}$

4.3 Leakage

According to the ACM0022 (Version 03.0), leakage emissions are associated with composting/co-composting, anaerobic digestion and the use of RDF that is exported outside the project boundary after June 2023.

And the waste by-products of the project do not involve compost or co-compost and the use of RDF outside the project boundary. Leakage emissions associated with anaerobic digestion of waste ($LE_{AD,y}$) are calculated according to "TOOL14: Project and leakage emissions from anaerobic digesters"(Version 02.0). According to the tool, the leakage emissions associated with the anaerobic digester ($LE_{AD,y}$) depend on how the digestate is managed. They include emissions associated with storage and composting of the digestate. Since the solid digestate will be outward transported and incinerated directly in thermal power plant and will not be stored under anaerobic conditions after June 2023 and the wastewater is treated aerobically by wastewater treatment plant nearby, leakage emission has been calculated until June 2023 and rest of the years of crediting period have been considered zero.

$\boxed{LE_{AD,y} = 2{,}737\ tCO_2e\ \text{(annual average)}}$

4.4 Net GHG Emission Reductions and Removals

To calculate the emission reductions the project participant shall apply the following equation:

$$ER_y = BE_y - PE_y - LE_y$$

Where:

ER_y	=	Emissions reductions in year *y* (t CO_2e)
BE_y	=	Baseline emissions in year *y* (t CO_2e)
PE_y	=	Project emissions in the year *y* (t CO_2e)

[18] https://webapi.teias.gov.tr/file/512cbf1d-0ca3-4492-b901-3722c7b682f7?download

LE_y = Leakage emissions in year y (t CO_2e)

Therefore, during the first crediting period, the total ex ante estimated emission reduction is 2,240,299 tCO2e, the average annual emission reduction is 320,042 tCO2e.

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
1	151,298	1,816	3,417	146,065
2	227,619	3,573	7,362	216,684
3	293,516	4,992	8,379	280,145
4	350,639	19,933	0	330,706
5	400,364	19,933	0	380,431
6	443,949	19,933	0	424,016
7	482,187	19,933	0	462,254
Total	2,349,572.13	90,115	19,158	2,240,299

5 MONITORING

Flow meters, sampling devices and gas analysers are subject to regular maintenance, testing and calibration to ensure accuracy. Parameters for monitoring under the ACM0022 Methodology.

5.1 Data and Parameters Available at Validation

Data / Parameter:	***$RATE_{COMPLIANCE,t}$***
Data unit	Fraction
Description	Rate of compliance with a regulatory requirement to implement the alternative waste treatment t implemented in the project activity
Source of data	Official studies, reports and certification from municipal authorities
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	To determine DF_{RATE}
Comments	Calculated based on the number of instances of compliance identified at the time of the investment decision and updated once for every subsequent crediting period

Data / Parameter:	**φdefault**
Data unit	-
Description	Default value for the model correction factor to account for model uncertainties
Source of data	"Methodological Tool: Emissions from solid waste disposal sites" (Version 08.0)
Value applied	0.85 (table below)
Justification of choice of data or description of measurement methods and procedures applied	Default value suggested in the tool. The project selects Option 1 :Use a default value to determine $φ_y$ in the tool "Emissions from solid waste disposal sites" (version 08.0), and the project belongs to the situation of Application B: The VCS project activity mitigates methane emissions from a specific existing SWDS, thus 0.85 is chosen for φ_y.
Purpose of Data	Calculation of baseline emissions
Comments	-

	Humid/wet conditions	**Dry conditions**
Application A	0.75	0.75
Application B	0.85	0.80

Data / Parameter:	DOCf,y
Data unit	Weight fraction
Description	Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposes in the SWDS
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	Using IPCC 2006 Guidelines default value
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter:	OX
Data unit	-
Description	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	0.1
Justification of choice of data or description of measurement methods and procedures applied	The landfill is covered with plastic film or soil, for conservative 0.1 was chosen.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter:	F
Data unit	-
Description	Fraction of methane in the digester gas (volume fraction)
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, Vol. 5 Waste,
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Ex ante calculation of baseline emissions
Comments	-

Data / Parameter:	MCF
Data unit	-
Description	Methane correction factor
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, Vol. 5 Waste, Chapter-3, Table 3.1
Value applied	1
Justification of choice of data or description of measurement methods and procedures applied	In case that the SWDS does not have a water table above the bottom of the SWDS and in case of application B, then select the applicable value from the following: (a) 1.0 for anaerobic managed solid waste disposal sites. These must have controlled placement of waste (i.e.waste directed to specific deposition areas, a degree of control of scavenging and a degree of control of fires) and will include at least one of the following: (i) cover material; (ii) mechanical compacting; or (iii) levelling of the waste; (b) 0.5 for semi-aerobic managed solid waste disposal sites. These must have controlled placement of waste and will include all of the following structures for introducing air to the waste layers: (i) permeable cover material; (ii) leachate drainage system; (iii) regulating pondage; and (iv) gas ventilation system; (c) 0.8 for unmanaged solid waste disposal sites – deep. This comprises all SWDS not meeting the criteria of managed SWDS and which have depths of greater than or equal to 5 meters; (d) 0.4 for unmanaged-shallow solid waste disposal sites or stockpiles that are considered SWDS. This comprises all SWDS not meeting the criteria of managed SWDS and which have depths of less than five meters. This includes stockpiles of solid waste that are considered SWDS (according to the definition given for a SWDS) Application B is applicable. The site of Istanbul is an anaerobic managed solid waste disposal site, as described under (a).
Purpose of Data	Ex ante calculation of baseline emissions
Comments	-

Data / Parameter:	DOCj
Data unit:	% (Wet waste)
Description:	Fraction of degradable organic carbon in the waste type j (weight fraction)
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, Vol. 5 Waste, Tables 2.4 and 2.5.

Value applied	The following values for different waste types have been applied as a default value:

Waste Type j	DOC_j (% wet waste)
Wood and wood products	43
Pulp, paper and cardboard (other than sludge)	40
Food, food waste, beverages and tobacco (other than sludge)	15
Textiles	24
Garden, park and yard waste	20
Glass, plastic, metal, other inert	0
Sewage sludge	5

Justification of choice of data or description of measurement methods and procedures applied	Using IPCC 2006 Guidelines default value
Purpose of Data	Ex ante calculation of baseline emissions
Comments	-

Data / Parameter:	**kj**
Data unit:	1/yr
Description:	Decay rate for the waste type *j*
Source of data	Source: https://www.mgm.gov.tr
Value applied	The following values for different waste types have been applied as a default value:

Type of Waste		Climate Zone			
		Boreal and Temperature (MAT ≤ 20 C)		Tropical (MAT > 20 C)	
		Dry (MAP/PET < 1)	Wet (MAP/PET > 1)	Dry (MAP < 1000 mm)	Moist and Wet (MAP >= 1000mm)
		Default	Default	Default	Default
Slowly degrading waste	Paper/textiles waste	0.04	0.06	0.045	0.07
	Wood/straw waste	0.02	0.03	0.025	0.035
Moderately degrading waste	Other (non-food) organic putrescible/ Garden and park waste	0.05	0.1	0.065	0.17
Rapidly degrading waste	Food waste/Sewage sludge	0.06	0.185	0.085	0.4
Bulk Waste		0.05	0.09	0.065	0.17

Justification of choice of data or description of measurement methods and	Climatic conditions for Istanbul where project site located in: MAT – mean annual temperature: 14.3 °C MAP – Mean annual precipitation: 1,176 mm PET – potential evapotranspiration: 1,000 mm (max-annual) **MAP/PET = 1.18 (wet)** Therefore, Boreal and Temperate (MAT≤20°C) and dry (MAP/PET >)1 conditions are observed in the project site.
Purpose of Data	Ex ante calculation of baseline emissions
Any comment:	

Data / Parameter:	**GWPCH4**
Data unit:	tCO2e/tCH4
Description:	Global warming potential of CH4
Source of data	IPCC 5th Assessment Report15
Value applied	28
Justification of choice of data or description of measurement methods and	-
Purpose of Data	Ex ante calculation of baseline emissions
Any comment:	-

Data / Parameter	**Factor leaked CH4**
Data unit	m³ biogas leaked/m³ biogas
Description	Biogas leaked form biogas unit
Source of data	-
Value applied	0.05
Justification of choice of data or description of measurement methods and procedures applied	According to the Biogas produce in the biogas unit
Purpose of Data	For the calculation of PE_{phy} leakage default as AMS.III-AO
Comments	-

Data / Parameter	**$EF_{CH4,default}$**
Data unit	t CH4 leaked / t CH4 produced
Description	Digesters with steel or lined concrete or fiberglass digesters and a gas holding system (egg shaped digesters) and monolithic construction; Default emission factor for the fraction of CH4 produced that leak from the anaerobic digester
Source of data	-
Value applied	0.028 / 0.05
Justification of choice of data or description of measurement methods and procedures applied	the default value corresponding to the type of digester used in the project activity.
Purpose of Data	-
Comments	Applicable to the step “Determination of project emissions of methane from the anaerobic digester”

Data / Parameter	$TDLj,y$
Data unit	-
Description	Average technical transmission and distribution losses for providing electricity to source j in year y
Source of data	Source: ANNUAL DEVELOPMENT OF ELECTRICITY GENERATION- CONSUMPTION AND LOSSES IN TURKEY (1993-2019) https://webapi.teias.gov.tr/file/512cbf1d-0ca3-4492-b901-3722c7b682f7?download
Value applied	0.11
Justification of choice of data or description of measurement methods and procedures applied	TEİAŞ is publicly owned transmission company which publishes official statistics about electricity generation and distribution.

Purpose of Data	Calculate Project Emissions
Comments	Due to project doesn't involve the emissions from consumption of fossil fuels, only emissions from use of electricity for the operation and electricity from grid has already been considered when calculation the quantity net electricity generation, project emissions from electricity consumption is considered 0.

Data / Parameter:	D_{CH4}
Data unit	t CH4/m^3 CH4
Description	Density of methane at the temperature and pressure of the landfill gas in the year y
Source of data	Default value as per the latest version of "Tool to determine project emissions from flaring gases containing methane"
Value applied	0.00067 at normal condition
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter:	$EF_{CO2,m,i,y\ (combined)}$
Data unit	tCO2/MWh
Description	CO2 emission factor of fuel type i in year y
Source of data	Turkish National Electricity Network pdf
Value applied	0.5706
Justification of choice of data or description of measurement methods and procedures applied	The Emission Factor is given from Turkish National Electricity Network calculated annually and for the year 2019
Purpose of Data	Calculation Emission Factor
Comment	-

Data / Parameter:	EG_y
Data unit	MWh
Description	Net electricity generated in the project electricity system in year
Source of data	Electricity Meter
Value applied	101,546

Justification of choice of data or description of measurement methods and procedures applied	The operating hours are monitor from the gas engine software and are monitor and saved in the internal files.
Purpose of Data	Calculation of Project emission
Comments	

Data / Parameter	**$\eta_{m,y}$**
Data unit	1/yr
Description	Average net energy conversion efficiency of power unit m in year y
Source of data	As per Factsheet of the gas engines
Value applied	0.43
Justification of choice of data or description of measurement methods and procedures applied	According to the electricity produce in ratio to the Methane as input
Purpose of Data	-
Comments	-

Data / Parameter	**$\rho CH4,n$**
Data unit	kg/m3
Description	Density of methane gas at reference conditions
Source of data	Default Value of TOOL06
Value applied	0.716
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Determination of the methane mass flow of the residual gas
Comments	-

Data / Parameter	***$SPEC_{flare}$***
Data unit	Temperature - °C Flow rate or heat flux - kg/h or m3/h Maintenance schedule - number of days
Description	The operator will decide whether to start and ignite the Flare with Cogeneration either on or off (see settings page on Data sheet specifications).
Source of data	Specialist in Biogas technology
Value applied	Due to almost it has been not used and for simplification, it is zero.
Justification of choice of data or description of measurement methods and procedures applied	*High temperature combustion and extraction system*
Purpose of Data	Calculation of project emissions from residual gas
Comments	Please find the specifications on the folder data sheet of equipment. This in an enclosed flare.

5.2 Data and Parameters Monitored

Parameters for monitoring under the ACM0022 Methodology.

Data / Parameter:	**$LFG_{total,y}$**
Data unit:	Nm3
Description:	Total amount of landfill gas captured under Normal Temperature and Pressure in year y
Source of data	Measured by flow meters. The data will be aggregated regularly (monthly-yearly)
Description of measurement methods and procedures to be applied	-

Frequency of monitoring/recording	Continuous measurement by the flowmeters in real time.
Value applied	To be monitored during verification
Monitoring equipment	Flow Meter
QA/QC procedures to be applied	Data is measured continuously with a flow meter by the Project owner. Flow meters will be subject to a regular maintenance and testing regime to ensure accuracy. This will ensure that the accuracy of the measurement instrument is maintained. The measurement frequency is equal to or more than one sampling each hour.
Purpose of data	
Calculation method	-
Comments	-

Data / Parameter:	$LFG_{electricity,y}$
Data unit:	Nm^3
Description:	Total amount of landfill gas combusted in the gas generator under normal temperature and pressure in year y
Source of data	Measured by flow meters. The data will be aggregated regularly (monthly-yearly) for each gas generator.
Description of measurement methods and procedures to be applied	LFG electricity measured via flowmeter.
Frequency of monitoring/recording	Continuous measurement by the flowmeters in real time.
Value applied	To be monitored during verification
Monitoring equipment	Flow Meter
QA/QC procedures to be applied	Data is measured continuously with a flow meter by the Project owner. Flow meters will be subject to a regular maintenance and testing regime to ensure accuracy. This will ensure that the accuracy of the measurement instrument is maintained. The measurement frequency is equal to or more than one sampling each hour.

Purpose of data	This parameter will be used for emission reduction calculation.
Calculation method	-
Comments	-

Data / Parameter:	$F_{CH4,RG,m}$
Data unit:	kg
Description:	Mass flow of methane in the residual gas in the minute m
Source of data	Flowmeter
Description of measurement methods and procedures to be applied	The flow rate of the residual landfill gas is automatically converted to dry basis and normal conditions by the PLC system without any intervention by the Project owner. This parameter is calculated by multiplying LFGflare, density of methane and wCH4 (percentage of methane measured by the gas analyzer). This is in compliance with the “Tool to determine the mass flow of a greenhouse gas in a gaseous stream”, v.03.0.0. For conversion, volumetric flow rate data will collected from the flow meter Nr.1 (FM 1).
Frequency of monitoring/recording	Measured continuously in real-time and recorded every 30 minutes.
Value applied	To be monitored during verification
Monitoring equipment	Flow meter
QA/QC procedures to be applied	A separate flow meter is installed for the flare unit. Data is measured continuously with a flow meter by the Project owner. Flow meters will be subject to a regular maintenance and testing regime to ensure accuracy. This will ensure that the accuracy of the measurement instrument is maintained. The measurement frequency is equal to or more than one sampling each hour.
Purpose of data	This parameter will be used to check with the $SPEC_{flare}$ parameter. This parameter will be used to calculate $PE_{flare,y}$
Calculation method	-
Comments	-

Data / Parameter:	w_{CH4}
Data unit:	%
Description:	Methane fraction in the landfill gas
Source of data	Gas analyser (at the main booster pipe)
Description of measurement methods and procedures to be applied	Methane content is measured directly and continuously with a gas analyser.
Frequency of monitoring/recording	Continuous measurement by the flowmeters in real time.
Value applied	53%
Monitoring equipment	-
QA/QC procedures to be applied	The gas analyser will be maintained and calibrated regularly according to the manufacturer's requirements in order to ensure that required level of accuracy is maintained. The gas analyser is subject to a regular maintenance and testing regime to ensure accuracy, therefore the analyser will be calibrated according to the manufacturer's recommendations. The measurement interval is equal to or more than one sampling each hour.
Purpose of data	The percentage of methane in the landfill gas is used to calculated the amount of methane destroyed in the gas engine and flare units. It is also used to check whether it satisfies the minimum methane content threshold for the flare unit to ensure the gas is flared efficiently.
Calculation method	-
Comments	-

Data / parameter:	**Operation of the power plant**
Data unit:	8,760
Description:	Operational duration of the power plant
Source of data:	Gas Engine System, Scada System

Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	Monitored daily, monthly and yearly
Value monitored	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of the data	Calculation of Baseline/project emissions from electricity generation
Calculation method	-

Data / parameter:	$T_{EG,m}$
Data unit:	°C
Description:	Temperature in the exhaust gas of the enclosed flare in minute m
Source of data:	Scada System
Description of measurement methods and procedures to be applied	If more than one temperature port is fitted to the flare, the flare manufacturer must provide written instructions detailing the conditions under which each location shall be used and the port most suitable for monitoring the operation of the flare according to manufacturers specifications for temperature
Frequency of monitoring/recording	Daily
Value monitored	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of the data	Calculation of project emissions from flaring

Calculation method	-
Comments	-

Data / parameter:	**ER_y**
Data unit:	t CO2e/y
Description:	Emission Reductions in year y
Source of data:	-
Description of measurement methods and procedures to be applied	If more than one temperature port is fitted to the flare, the flare manufacturer must provide written instructions detailing the conditions under which each location shall be used and the port most suitable for monitoring the operation of the flare according to manufacturer's specifications for temperature
Frequency of monitoring/recording	Once for each monitoring period
Value monitored	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of the data	-
Calculation method	-
Comments	-

Data / Parameter:	$EGPJ,y$
Data unit:	MWh/yr
Description:	Quantity of net electricity (difference between the total quantity of electricity generated by the power plant/unit and the auxiliary electricity consumption) generation that is produced and fed into the grid as a result of the implementation of the project activity in year y
Source of data	Meter reading protocols (OSOS records)

Description of measurement methods and procedures to be applied	The quantity of net electricity generation supplied to the grid by the project plant will recorded and reported based on the DEDAS meters at the project site.
Frequency of monitoring/recording	Continuous measurement and at least monthly recording. (Remote automatic meter reading system-OSOS)
Value applied	101,546
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	The data will be cross checked with sales receipts. The electricity meter will be periodically calibrated
Purpose of data	Calculation of baseline emissions
Calculation method	Continuous measurement and at least monthly recording. (Remote automatic meter reading system-OSOS)
Comments	-

Data / parameter:	W_x
Data unit:	tons
Description:	Total amount of waste disposed in a SWDS in year x
Source of data:	Weighbridge data
Description of measurement methods and procedures to be applied	The garbage truck which carries municipal waste is weighed and printed receipts are filed
Frequency of monitoring/recording	The total amount of waste is continuously measured and the data is aggregated at least annually.
Value monitored	2,000 t/day
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of the data	Baseline Emissions from SWDS
Calculation method	-

Comments	-

Data / parameter:	$\eta_{flare,m}$
Data unit:	% (Percentage)
Description:	Flare efficiency
Source of data:	Default Value of TOOL06
Description of measurement methods and procedures to be applied	Since option A conditions are met and enclosed flare could be defined low height, 80% has been chosen.
Frequency of monitoring/recording	Continuously
Value monitored	80%
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of the data	Determination of project emissions from flaring
Calculation method	-
Comments	

Data / Parameter:	$NCV_{i,y}$
Data unit:	GJ per mass or volume unit (e.g. GJ/m³, GJ/ton)
Description:	Weighted average net calorific value of fuel type i in year y
Source of data:	Measurements by the project proponent
Description of measurement methods and procedures to be applied	Measurements be undertaken in line with national or international fuel standards

Frequency of monitoring/recording	The NCV be obtained for each fuel delivery, from which weighted average annual values be calculated.
Value monitored	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of the data	-
Calculation method	-
Comments	-

Data / Parameter:	$FC_{H4,EG,t}$
Data unit:	kg
Description:	Mass flow of methane in the exhaust gas of the flare on a dry basis at reference conditions in the time period t
Source of data:	Measurements undertaken by a third-party accredited entity
Description of measurement methods and procedures to be applied	Values taken form the system that has been calibrated
Frequency of monitoring/recording	Biannual
Value monitored	-
Monitoring equipment	-
QA/QC procedures to be applied	According to standard applied
Purpose of the data	-
Calculation method	-
Comments	The project implements an enclosed flare therefore, there is monitoring.

Data / Parameter:	$V_{i,RG,m}$

Data unit:	-
Description:	Volumetric fraction of component i in the residual gas on a dry basis in the minute m where i = CH4,
Source of data	Measurements by project operator using a continuous gas analyzer (values are recorded with the same frequency as the flow).
Description of measurement methods and procedures to be applied	Measurement may be made on dry basis.
Frequency of monitoring/recording	Continuously. Values are to be averaged on a minute basis if flare is on operation 100% status
Value monitored	-
Monitoring equipment	-
QA/QC procedures to be applied	Analyzers must be periodically calibrated according to the manufacturer's recommendation.
Purpose of the data	-
Calculation method	-
Comments	Project operator measure the content CH4, CO2

Data / Parameter:	$v_{RG,m}$
Data unit:	m3
Description:	Volumetric flow of the residual gas on a dry basis at reference conditions in the minute m
Source of data:	Measurements by project participants using a flow meter
Description of measurement methods and procedures to be applied	Measurements by project operator using a calibrated flow meter
Frequency of monitoring/recording	Continuously. In flare operation is 100% or on.
Value monitored	-

Monitoring equipment	-
QA/QC procedures to be applied	Flow meters are to be periodically calibrated according to the manufacturer's recommendation
Purpose of the data	-
Calculation method	-
Comments	This parameter is monitored, project implements enclosed flare

Data / Parameter:	$fc_{CH4,EG,m}$
Data unit:	mg/m3
Description:	Concentration of methane in the exhaust gas of the flare on a dry basis at reference conditions in the minute m
Source of data	Measurements by project operator using a continuous gas analyser
Description of measurement methods and procedures to be applied	Extractive sampling analysers with water and particulates removal
Frequency of monitoring/recording	Continuously
Value monitored	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of the data	-
Calculation method	-
Comments	-

5.3 Monitoring Plan

The responsible entity for the monitoring system is Vega-Hereko personnel. The monitoring activities will primarily involve four types of personnel: the Operation Site Manager, Field Operator, Electrical & Mechanical Operator and the Laboratory Operator.

The Field Operators will perform activities such as monitoring and checking operations of the blower and flare, recording data at the blower/flare station, routine maintenance of collection system components, preparing daily logs and completing checklists, and send data to the Operation Site Manager.

The Operation Manager's responsibilities include reviewing the data collected both manually by the Field Technicians and the one recorded automatically by analytical equipment, making recommendations and/or implementing system adjustments to maximize methane capture and destruction, scheduling monitoring and O&M activities, performing quality assurance checks on operations, coordinating with system component manufacturers as needed, to maintain proper operations and calibration, and compiling data as required by the Methodology.

Only for manual data collection, the Operation Manager will be responsible for reviewing the data collected.

Project Management Responsibility

The operators of the Methane recovery and electricity generation system will be responsible for collecting all data monitored on-site. The operating and maintenance personnel will be skilled technicians, with extensive experience in equipment operation, maintenance and calibration, and emergency procedures. Overall responsibility for the monitoring and maintenance of all required tasks and their adequate management lie with the project manager. Detailed roles and responsibilities of the relevant staff involved in VCS monitoring are placed.

Training of Monitoring Personnel

The monitoring personnel is trained in the beginning of the project; the purpose of this training is to operate the project in a well manner. Periodical training will be defined by Vega-Hereko.

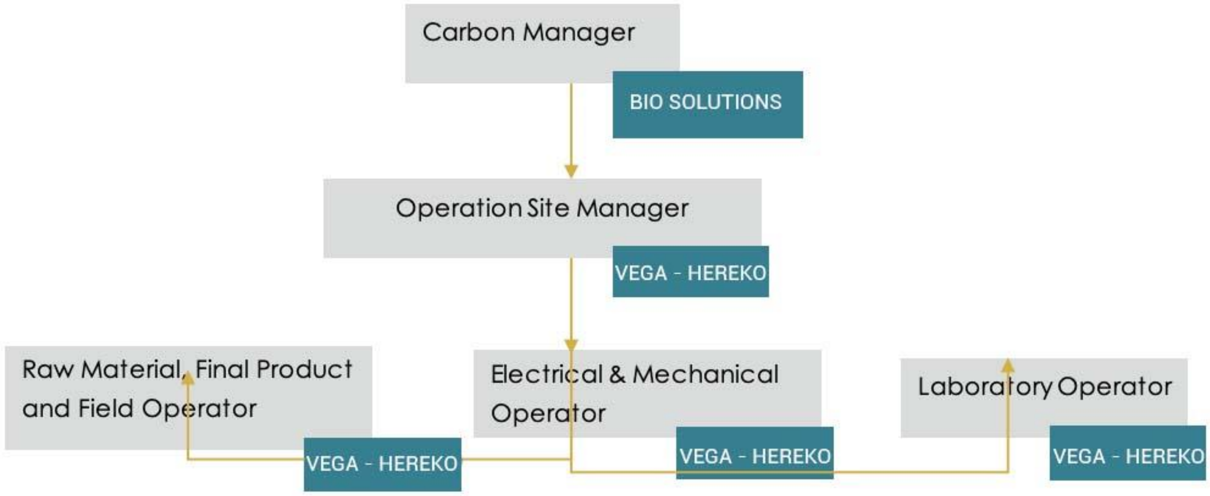

Figure 12. Organizational Structure of the Carbon Project

Data Analysis

The collected data will be reviewed and analyzed on a daily basis by the Operation Manager. In case of a drift of one parameter the Manager or Technician can react quickly to fix potential problems. All data required for the emission reduction calculations will be kept in the onsite-monitoring database. The Monitoring Manager will be the responsible to report the necessary information to Vega-Hereko.

The quality actions that will guarantee the success of the monitoring plan are the following:

Maintenance Plan

The plant's computerized management and control system includes: the management of level management in gas meters through the level indicators; monitoring and control of the heating system; alarm management of minimum and maximum levels and security; and monitoring of the data analysis of the gas analysis. With regards to the electricity generation component of the project, given that it dispatches electricity to the National Grid, measuring, recording, storing, aggregating, collating and reporting data and parameters will follow the procedure described by the authorities.

The following aspects are core to the maintenance of the monitoring system in order to assure proper data monitoring during the project:

- Equipment preventive maintenance
- Equipment calibration

All data collected as part of monitoring will be archived electronically and be kept at least for 2 years after the end of the last crediting period. Regarding the monitoring equipment of the methane recovery component of the project, maintenance and calibration will be performed in line with manufacturers' recommendations. With regards to the electricity meter, it is a high accuracy measurement device/s and meets all relevant metrological requirements prescribed by the state authority. Procedures for maintenance of the meter will be conducted in accordance with national procedures and standards.

Quality assurance and Control (QA/QC)

Quality control and quality assurance procedures will guarantee the quality of monitored data.

All data is archived electronically, backed up regularly and kept at least for 2 years after the end of the last crediting period. There are corrective actions when the parameters are out of the permitted range.

Regarding the monitoring equipment of the methane recovery component of the project, maintenance and calibration will be performed in line with manufacturers' recommendations. With regards to the electricity meter, as mentioned above, it is a high accuracy measurement device/s and meets all relevant metrological requirements prescribed by the state authority. Procedures for maintenance of the meter will be conducted in accordance with national procedures and standards.

Flaring

For the control of the Flare ignition, all management and operation is trough PLC, on screen looks like a switch. The user will decide whether to start and ignite the Flare with Cogeneration either on or off. This is all managed with secure data logging and storage of data is happening automatically to be downloaded in excel.

Calibration plans:

The company SICK SENSOR performs every year calibration visits from the gas flow meters in the booster unit, the values of O2%, CO2% and CH4% are measured. Please see the calibration certificates documents.

The gas engines have their own monitoring system and parameters can be seen every time on the display. There is an internal storage of the data on PC and on the cloud. All data can be downloaded in an excel file.

Emergencies Procedures

The running hours of the power plants will be monitored as part of the monitoring procedures. In case of failure of one of the monitoring devices, portable instruments will be used in order to carry out periodic daily monitoring of the missing parameter(s). These data are recorded on paper. Vega-Hereko defined emergency procedures according to the provider recommendations.

APPENDIX

Not Applicable.